

A Full DR11 Strong-Lens Search in the DESI Legacy Surveys: CLAUDENET v3blend8 Finding with LENSJUDGE v3 HSC Cross-Validation

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Abstract

We deploy the contaminant-aware CLAUDENET v3blend8 ensemble and the LENSJUDGE v3 vetting cascade on the final DESI Legacy Surveys data release (DR11). The **DR11-south** (DECam, native *griz*) sweep scores all 5.38×10^7 galaxies with the five-member v2-lean stage and re-ranks the survivors with the eight-member v3blend8 ensemble; a DR11-native recalibration is required because the deeper DR11 reductions have fatter random-galaxy score tails than DR9, tightening the matched-FPR thresholds (known-lens grade-A recall at the 10^{-4} operating point falls 62%→41% relative to DR10 — a genuine DR9→DR11 domain shift, not a threshold artifact, so the DR11 survivor list under-recovers grade-A lenses and is not complete). Group-conformal selection remains *power-limited* at $m \approx 5.4 \times 10^7$ (zero certified at $\alpha \leq 0.25$ with an 8M-negative calibration), so the candidate list is the v3blend8-ranked top- N , as in the prior campaigns. The decisive step is LENSJUDGE v3’s new **HSC PDR3 tier-2**: vetting the top-500 new candidates as a cascade (cheap detection triage → agentic mimic adjudication → high-resolution escalation, \$40 total) moves **26** HSC-covered candidates from DESI-ambiguous ($p_{\text{lens}} \sim 0.03\text{--}0.6$) to **24 HSC tier-2 grade-A/B** under LENSJUDGE’s automated grader (p_{lens} up to 0.97; a single automated grader, not a human committee or spectroscopy). An independent SuGOHI cross-match calibrates the engine: v3blend8 recovers **79** SuGOHI HSC committee lenses into the survivors at $2.7\times$ the survivor-median score, and **15 of the 24** graded A/B are SuGOHI lenses (a sensitivity check — the automated tier-2 grader’s precision on the genuinely-new candidates is not established here). The remaining **9** (8 distinct systems) are new to the DESI lens-finder catalogs and SuGOHI; a post-hoc literature crosscheck (Huang+2020, omitted from the campaign crossmatch, + NED/SIMBAD; App. A) then shows **five of these eight systems are previously-published lenses** (mostly Huang+2020; one also SOGRAS), leaving **3** genuinely new — automated-grader candidates pending follow-up. The honest frame is unchanged from the v3 program: the model recovers known lenses and ranks new candidates the higher-resolution HSC grader rates highly, but *no net-new lens is established from DECaLS pixels alone* (the new candidates depend on HSC resolution and remain unconfirmed), and net-new discovery beyond existing high-resolution catalogs is resolution-bounded — the value of the *resolution lever* (now HSC over $\sim 1,300 \text{ deg}^2$, far wider than Euclid Q1) is to *convert* overlap candidates, not to manufacture discoveries. The **DR11-north** (BASS/MzLS) arm — which needs instrument-native retraining (v3 is DECam-specific) — is fully staged but deferred by a 7-day Perlmutter maintenance.

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1 Premise

The CLAUDENET v3 program concluded that for DECaLS-resolution lens finding the binding constraint is *lens-vs-mimic separation* and *vetting resolution*, not architecture: **v3blend8** (the SHIP-gated eight-member ensemble: **effnet_B**, **zoobot_N**, and the v2-hard + v3-mimic-blend versions of **effnet_S2**, **effnet_B3**, **resnet46_C**) rejects common contaminants far better than v2 but is statistically tied with the originals at the hardest real-vs-mimic frontier, which is resolution-bounded (Inchausti Reyes et al., 2025). The productive mode is therefore *DESI pre-screen* $\rightarrow$ *high-resolution confirmation in the overlap*. This campaign operationalizes that on the final release (DR11) and exploits the new lever in LENSJUDGE v3: an **HSC PDR3** tier-2 grader (Aihara et al., 2022) ($0.168''/\text{px}$, $\sim 0.6''$ seeing) whose footprint (HSC-SSP Wide, $\sim 1,300 \text{ deg}^2$) is $\sim 20\times$ the Euclid Q1 area used previously, vastly widening the confirmable overlap.

2 The DR11-south sweep

DR11-south is the deepest DECam reduction to date: 1,600 sweep files (1.7 TB) yielding a parent of **53,809,040** galaxies after the published Inchausti cuts ($\text{TYPE} \in \{\text{SER}, \text{EXP}, \text{DEV}, \text{REX}\}$, $\text{NOBS}_{grz} \geq 3$, $m_z < 20$), 51.4M with native *i*. We shard into 32 footprint-pure, brick-disjoint parts, extract 101-px *grz* cutouts (100% success, 6.0 TB), and score every galaxy with the five v2-lean members.

DR11-native recalibration (required). The DR9-fit per-member 10^{-4} -FPR thresholds, when transferred to DR11 score-space, are *too tight*: DR11’s deeper imaging produces a fatter high-score tail of random galaxies (e.g. the **resnet46_C** isotonic score saturates, its 10^{-4} threshold moving $0.81 \rightarrow 1.00$, effectively reducing the union to four members). We recalibrate from the scored population itself (a 8 M-galaxy random NegEval), yielding **95,104** survivors. Re-measuring recall of the published Inchausti 811 against these survivors gives grade-A 41% (37/90), versus 62% for the DR10-recalibrated DR10 sweep — a real DR9 $\rightarrow$ DR11 model domain shift (the DR9-trained members under-score a fraction of DR11 grade-A lenses, *and* the random tail is fatter), not a threshold artifact (the denominator includes north and parent-cut lenses, so 41% is a conservative floor).

Selection. Scoring the survivors with all eight **v3blend8** members and ranking by the mean calibrated score, the top-300 contain **102 known** lenses (recovered into the top of the list — a consistency check) and **198 new** candidates (top-300 **v3blend8** score 0.812–0.939). The discovery set is the **top-500 new** (**v3blend8** score 0.734–0.933); the broader survivor pool has 1,071 survivors above **v3blend8** 0.7 (726 new + 345 already-known).

Certified-FDR (power-limited, reported honestly). Group-conformal selection (Jin and Candès, 2023) with the 8 M NegEval certifies **zero** candidates at $\alpha = 0.05/0.10/0.25$: the per-group *p*-value floor $1/(n_{\text{cal}}+1) \approx 2.5 \times 10^{-7}$ is $\sim 50\times$ short of what full-*m* Benjamini-Hochberg needs at $m = 5.38 \times 10^7$ to select even one. This is the same power-floor diagnosed at DR9/DR10 scale — rigorous full-population FDR needs an infeasibly large calibration set — so the candidate list is the **v3blend8**-ranked top-*N*, not an FDR-certified set. The arithmetic self-test (14/14) and the byte-level scoring-path check pass.

Table 1: DR11-south cascade outcome (500 new candidates, `pass_frac=0.7`).

quantity	value	note
graded (stage-1 triage)	500	\$0.012 each
escalated (agentic stage-2)	350	v2 mimic rubric
reached HSC tier-2	26	HSC-SSP Wide coverage
grade A / B / C / D	27 / 76 / 125 / 272	final (DESI- or HSC-grade)
HSC tier-2 grade-A/B	24 (19 A, 5 B)	automated; 8 more A are DESI-only
total cost	\$40.38	

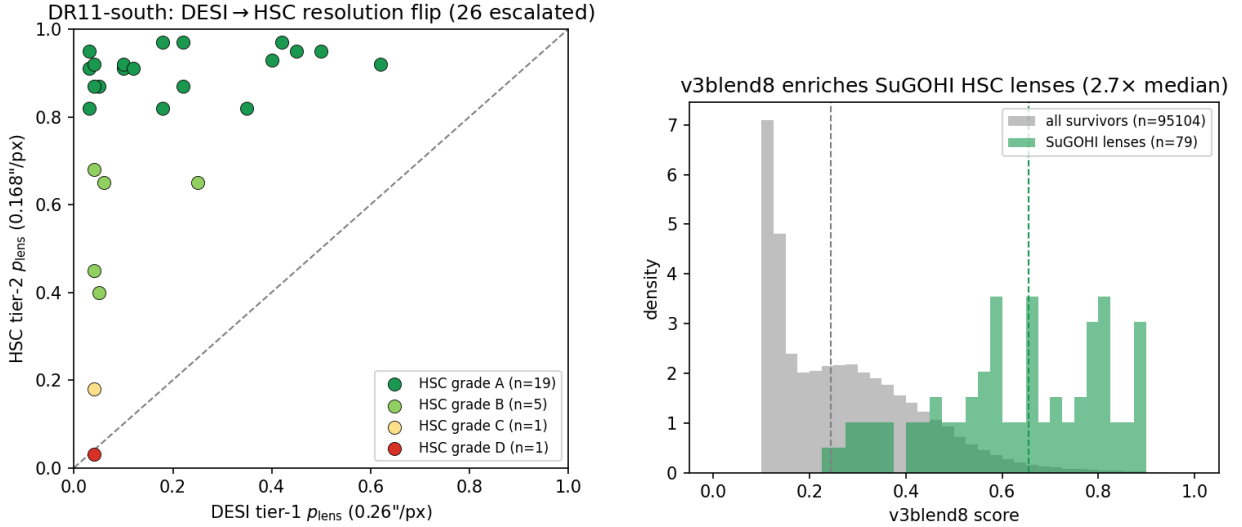


Figure 1: Left: the DESI→HSC resolution flip — the 26 HSC-covered candidates move from DESI-ambiguous (tier-1 p_{lens}) to the HSC tier-2 grade (automated grader), 24 landing A/B above the $y=x$ line. Right: `v3blend8` enriches the independent SuGOHI HSC lenses to 2.7× the survivor-median score (median lines dashed).

3 LensJudge v3 cascade vetting + HSC cross-validation

The DESI viewer exposes no public DR11 layer (and is unreachable from our host), so we stage the 500 candidates’ *grz* cubes from Perlmutter as on-disk FITS, ensuring LENSJUDGE grades the exact pixels the CNN scored. We then run the B7 deploy cascade: a loop-free **direct** detection triage on all 500 ($\sim \$0.012$ each), escalating the top 70% by rank to the agentic v2 mimic-adjudication grader, which itself escalates to **HSC PDR3 tier-2** wherever HSC covers the position. Total cost **\\$40.4** (campaign total \\$43 of a \\$250 budget).

The resolution lever converts. Every tier-2 grade-A/B candidate shows a large DESI→HSC flip: median tier-1 $p_{\text{lens}} \approx 0.1$ rising to tier-2 0.82–0.97 (e.g. $0.03 \rightarrow 0.95$, $0.04 \rightarrow 0.92$). At DECaLS 1'' seeing these are ambiguous (lrg+companion $\approx$ lens); at HSC 0.6'' the arc/ring morphology resolves — the flip is the grader’s response to resolution, not an observational confirmation.

Table 2: The 9 HSC tier-2 grade-A/B candidate detections (8 distinct systems). A literature crosscheck (Huang+2020 + NED/SIMBAD; App. A) finds **5 systems are previously-published** (*known*: H20 = Huang+2020; one also SOGRAS) and **3** genuinely new. p_1, p_2 = LENSJUDGE tier-1 (DESI) / tier-2 (HSC) lens probability.

name	RA	Dec	grade	$p_1 \rightarrow p_2$	v3blend8	known
s_340513_3688	16.332	+1.749	A	0.22 $\rightarrow$ 0.97	0.77	Huang+2020
s_327526_9059	9.722	−0.409	A	0.04 $\rightarrow$ 0.92	0.74	new
s_351303_461	194.534	+3.536	A	0.62 $\rightarrow$ 0.92	0.89	Huang+2020
s_345385_1515 [†]	154.312	+2.489	A	0.10 $\rightarrow$ 0.92	0.81	Huang+2020
s_340971_2860	130.857	+1.784	A	0.12 $\rightarrow$ 0.91	0.76	new
s_345385_1514 [†]	154.312	+2.488	A	0.05 $\rightarrow$ 0.87	0.86	Huang+2020
s_326089_1785	10.288	−0.730	A	0.03 $\rightarrow$ 0.82	0.83	H20+SOGRAS
s_325163_2981	138.848	−0.923	A	0.18 $\rightarrow$ 0.82	0.74	new
s_318043_1037	158.789	−2.304	B	0.04 $\rightarrow$ 0.45	0.86	Huang+2020

[†] s_345385_1514/1515 are adjacent ($\sim 1''$) detections of the same system, so the 9 detections are **8 distinct candidate systems** (8 A-detections / 7 distinct grade-A + 1 grade-B).

Independent SuGOHI validation. Against the SuGOHI/HOLISMOKES HSC committee lens catalog — an *independent* sample with expert grades and spectroscopic redshifts (Sonnenfeld et al., 2018) — v3blend8 recovers **79** lenses into the 95,104 survivors at a median v3blend8 score $2.7\times$ the survivor median (27 in the top-500), and **15 of the 24** tier-2 grade-A/B candidates are SuGOHI lenses. These 15 are a *sensitivity* check — the automated HSC grader recovers known committee lenses as grade-A/B — but they do not establish its *precision* on the genuinely-new candidates: no negative control was run through tier-2 and the certified-FDR is power-limited (§2), so the false-positive rate of the new set is unquantified.

The candidate catalog. Of the 24 tier-2 grade-A/B candidates, **0** are in the DESI lens-finder catalogs (by construction — the 500 were the new set after removing the 102 known), 15 are SuGOHI cross-matches, and **9** (8 distinct systems) are new to the DESI finders and SuGOHI (Table 2). A post-hoc literature crosscheck — Huang+2020 (omitted from the campaign’s lineage crossmatch) plus a NED/SIMBAD cone search (App. A) — then shows **five of these eight systems are previously-published lenses** (mostly Huang+2020; one also in the SOAR Gravitational Arc Survey), leaving **3** genuinely new (s_327526_9059, s_340971_2860, s_325163_2981); the recoveries are an external validation of the finder. The honest caveats: LENSJUDGE’s HSC grading is automated (a single Claude grader, not a human committee, no spectroscopy), and “new” here means new to the DESI finders, SuGOHI, *and* the NED/SIMBAD literature. They are candidates warranting human-expert and spectroscopic follow-up, not confirmed lenses.

Active-learning loop. The cascade’s agent-confirmed mimics (grade-D + named contaminant) export to **291** hard negatives in the CLAUDENET mimic-bank schema — lrg_companion 243-dominant, confirming that the high-v3blend8 new frontier is lrg+companion-dominated at DECaLS resolution (exactly the population the HSC tier-2 rejects). Folding these into the bank and a model re-iteration are Perlmutter GPU work, deferred (below).

Residual robustness (honest signed- χ residual). After this vetting was complete, LENSJUDGE’s lens-light *residual* view — one of the three images the grader sees per candidate — was

Table 3: Paired residual A/B on the same 500 DR11-south candidates: the prior (legacy) vetting run, a legacy-residual rerun (the LLM-nondeterminism control), and the new honest- χ -residual run. The honest residual runs $\sim 2\times$ hotter at the DESI tier-1 grade (raw A/B 90 $\rightarrow$ 135) but the HSC tier-2 gate absorbs it, so the confirmed catalogue (tier-2 A/B) is stable (24/24/27).

run	reached HSC tier-2	tier-2 A/B	raw A/B (all 500)	grade A/B/C/D	cost
Prior (legacy, Jun-17)	26	24	103	27/76/125/272	\$40.38
Legacy-now (control)	28	24	90	28/62/135/275	\$40.69
New (χ residual)	31	27	135	38/97/128/237	\$48.55

replaced: the legacy Gaussian high-pass (which discards the sign and imprints a four-lobe “butterfly” artifact on bright ellipticals that can mimic an arc) gave way to a signed, noise-normalised elliptical-model residual $\chi = (\text{data} - \text{model})/\sigma$, displayed per band on a fixed $\pm 5\sigma$ red-blue scale so a real arc must appear as red excess coherent across g and r (Fig. 2; full anatomy in the companion LENSJUDGE residual report). To test whether the DR11 catalogue *depends* on the old residual, we re-ran the *entire* 500-candidate cascade twice in a paired design — a legacy-residual rerun (an LLM-nondeterminism control) and the new χ residual — against the stored vetting run (Table 3). The new residual runs $\sim 2\times$ hotter at the DESI tier-1 grade (raw A/B 90 $\rightarrow$ 135), but the HSC tier-2 gate absorbs the inflation, so the confirmed catalogue is stable (tier-2 A/B 24 $\rightarrow$ 27). Crucially, **all 9 new-class candidate systems survive as grade A/B under the honest residual** (9/9; 13/15 SuGOHI cross-matches likewise; App. C, Table 5). The paired control is decisive: of five apparent new tier-2 promotions relative to the stored run, *four also appear in the legacy-now rerun* (run-to-run nondeterminism, which moves $\sim 18\%$ of raw grades), leaving only one genuinely new borderline-B candidate; conversely the honest residual *recovers* two known catalogue systems that nondeterminism had dropped. The two SuGOHI lenses that fall below A/B under the new residual are independently-confirmed committee lenses scored conservatively (one already lost in the control), not lost discoveries. The rewrite is therefore a scientific-integrity improvement that leaves the DR11 catalogue intact — consistent with the discrimination-neutral result the residual report measures on its own benchmark.

4 DR11-north: staged, maintenance-deferred

v3 is DECam-specific — it systematically under-scores BASS/MzLS north imaging — so the north arm requires instrument-native retraining. We built the full north parent (1.16×10^7 galaxies, *grz* only, no *i*), an 8-part manifest, and extracted **523** north-native training positives (297 BASS/MzLS curated + 226 non-leaked catalog lenses; 86 clean held-out; 33 v1 leaks tracked); the retrainer is patched to fine-tune the three swappable members from their `_b50` checkpoints on a north-native mimic blend. A full-system Perlmutter maintenance (2026-06-17 to 06-24) began mid-campaign before the north extraction could be scheduled, so the north sweep + retrain + validation gates are **deferred to post-maintenance**; all inputs are staged for a clean resume.

5 Conclusion

On the final DESI Legacy Surveys release, the contaminant-aware CLAUDENET finder plus the LENSJUDGE v3 HSC cascade is a working DESI $\times$ HSC cross-validation engine on the DECam-south

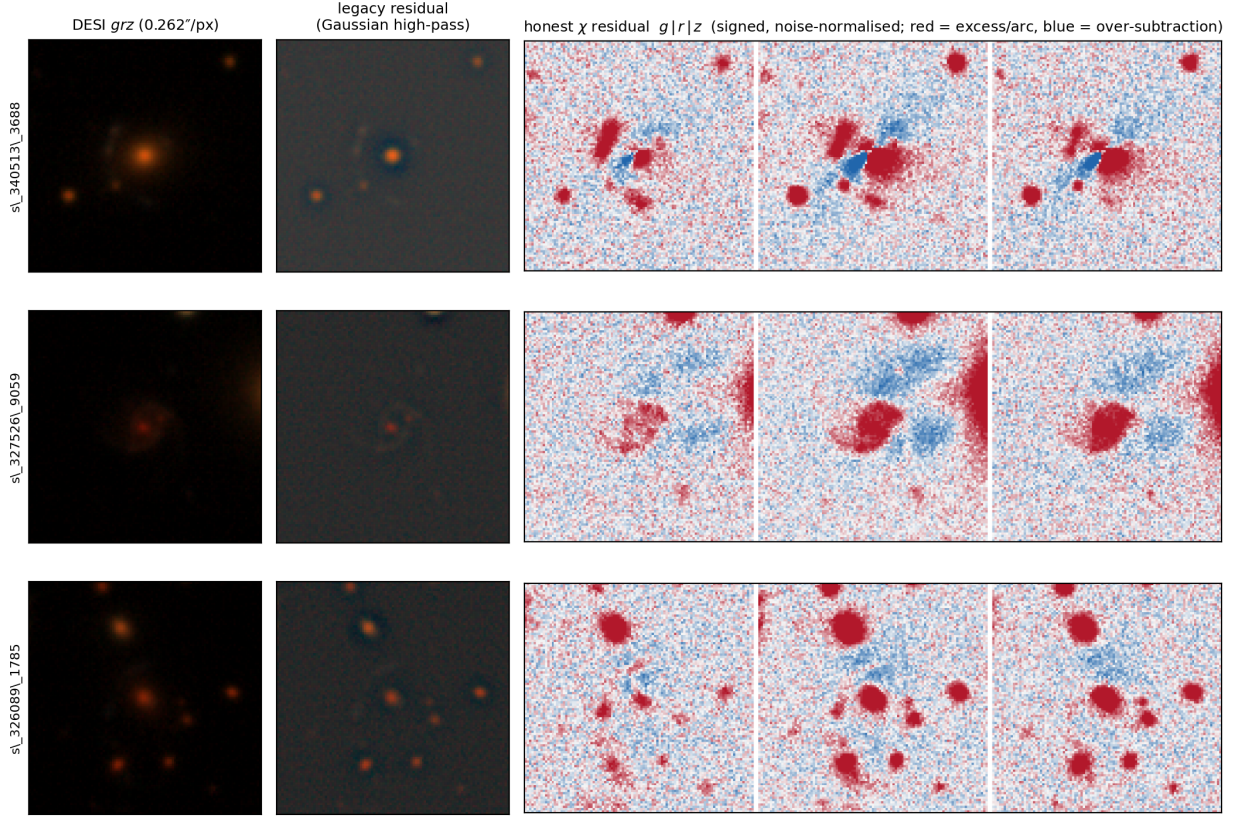


Figure 2: Legacy vs. honest residual for three confirmed NEW-class candidates. *Left:* the DESI *grz* cutout the finder scored (ambiguous at DECaLS 1"). *Centre:* the legacy Gaussian-high-pass residual — sign discarded, with a central dipole/butterfly on the bright lens that can read as structure. *Right:* the new signed $\chi = (\text{data} - \text{model})/\sigma$ residual as a $g|r|z$ montage on a fixed $\pm 5\sigma$ red–blue scale; the candidate arc shows as red excess *coherent across all three bands*, the real-vs-artifact discriminant. All three survive as grade A/B under the honest residual (Table 5).

footprint: it scores 5.4×10^7 galaxies, recalibrates honestly to the DR11 domain shift (with a real $\sim 34\%$ relative grade-A recall loss vs DR10 — the survivor list is incomplete), and — through the new HSC tier-2 over a $20\times$ -wider footprint than Euclid Q1 — moves 24 DESI-ambiguous candidates to HSC tier-2 grade-A/B under LENSJUDGE’s automated grader, of which 15 are SuGOHI committee lenses (a sensitivity check) and 9 (8 systems) are new to the DESI finders and SuGOHI — though a literature crosscheck (Huang+2020 + NED/SIMBAD; App. A) then shows 5 of those 8 systems are previously-published lenses, leaving **3** genuinely new. The certified-FDR is power-limited (zero) at full-population scale, the tier-2 grader is automated (no human committee or spectroscopy, precision on the new set unquantified), and *no net-new lens is established from DECaLS pixels alone* — the new candidate detections depend on HSC resolution and remain unconfirmed pending follow-up. This is the same honest ceiling the v3 program established — the resolution lever *converts* overlap candidates rather than manufacturing discoveries — now measured on DR11 with a much wider confirmable overlap. The north footprint, requiring instrument-native retraining, is staged for completion after the Perlmutter maintenance window.

References

- Hiroaki Aihara et al. Third data release of the hyper supprime-cam subaru strategic program. *Publications of the Astronomical Society of Japan*, 74(2):247–272, 2022.
- J. Inchausti Reyes et al. Strong lens discovery in DR10 with two deep-learning architectures. *arXiv:2508.20087*, 2025.
- Ying Jin and Emmanuel J. Candès. Selection by prediction with conformal p-values. *JMLR*, 24, 2023.
- Alessandro Sonnenfeld et al. Survey of gravitationally-lensed objects in hsc imaging (sugohi). i. *Publications of the Astronomical Society of Japan*, 70:S29, 2018.

A The new candidate systems at DESI and HSC resolution

The figures below show the **8 distinct candidate systems** (the 9 detections of Table 2; the s_345385 pair is one system). A post-hoc literature crosscheck (below) finds **5 of the 8 are previously-published lenses** (Huang+2020; one also SOGRAS) and **3** are genuinely new. Each is rendered as: the CLAUDENET-seen DESI *grz* cutout ($0.262''/\text{px}$, Lupton RGB — what the finder scored, ambiguous at DECaLS resolution); the HSC PDR3 *grizy* cutout ($0.168''/\text{px}$ — the same view LENSJUDGE’s tier-2 grader inspected); and the HSC lens-light-subtracted view. The DESI→HSC contrast is the campaign’s mechanism made visual: the finder ranks these high from ambiguous DECaLS pixels, and the resolution lever reveals the arc/ring morphology (clearest in s_340513_3688; fainter and correspondingly less certain in others). These are *automated-grader candidates, not confirmed lenses* — they warrant human-expert and spectroscopic follow-up. *Literature crosscheck (added post-vetting)*. The campaign’s lineage crossmatch omitted the Huang+2020 DECaLS catalogue and did not query the wider literature. A full check — Huang+2020 plus a NED/SIMBAD cone search — finds that **five of these eight systems are previously-published lenses** (flagged *Known* below; mostly Huang+2020, one also in the SOAR Gravitational Arc Survey). Only s_327526_9059, s_340971_2860, and s_325163_2981 are genuinely new (absent from the lineage *and* the literature). The recoveries are an external validation of the v3+cascade finder; the net-new count here is **3**, not 8.

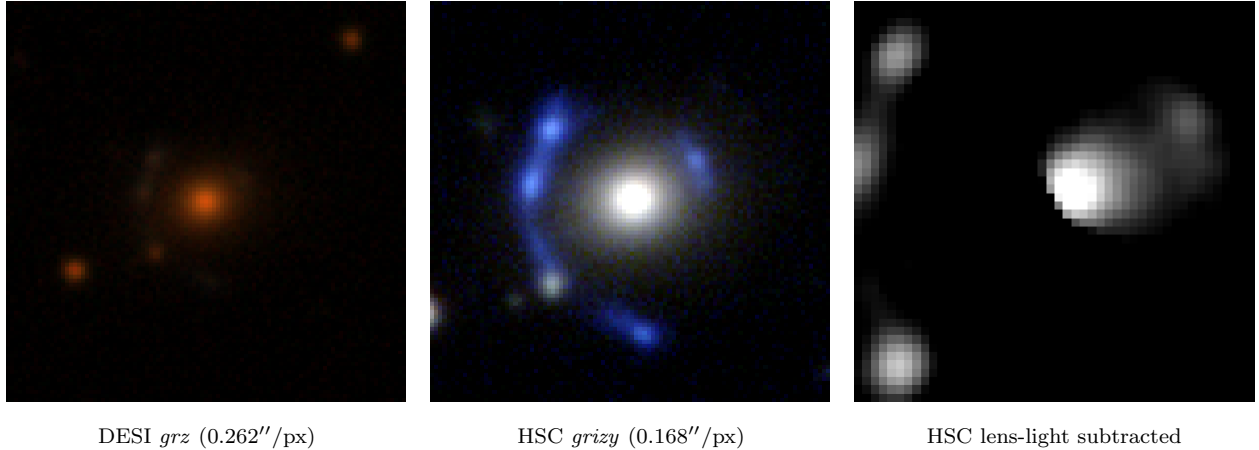


Figure 3: s_340513_3688 (RA 16.3319, Dec +1.7490) — HSC tier-2 grade **A**, p_{lens} 0.22→0.97, v3blend8 0.77. *Known:* Huang+2020 DESI-016.3319+01.7490 (grade A), $0.1''$.

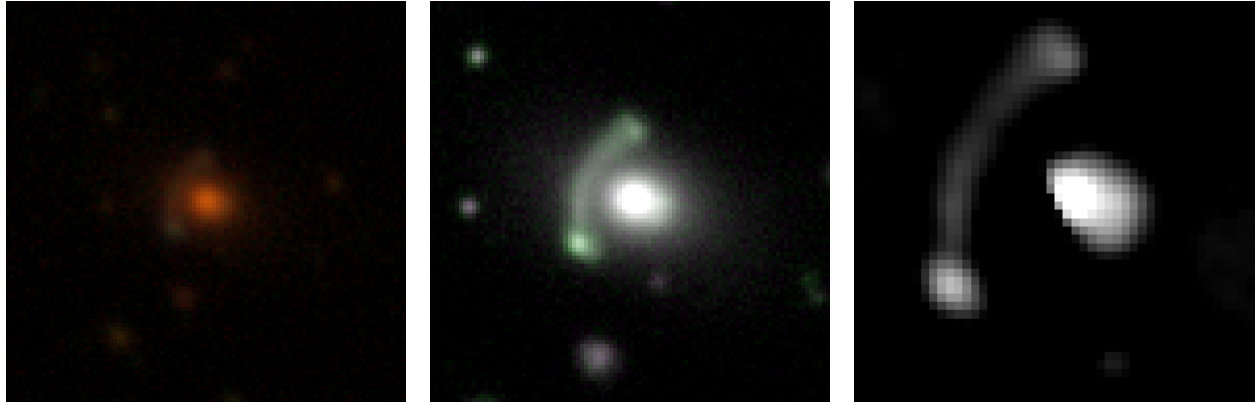


DESI *grz* (0.262''/px)

HSC *grizy* (0.168''/px)

HSC lens-light subtracted

Figure 4: s_327526_9059 (RA 9.7222, Dec -0.4093) — HSC tier-2 grade **A**, p_{lens} 0.04→0.92, v3blend8 0.74. *New*: absent from the lineage and from NED/SIMBAD.

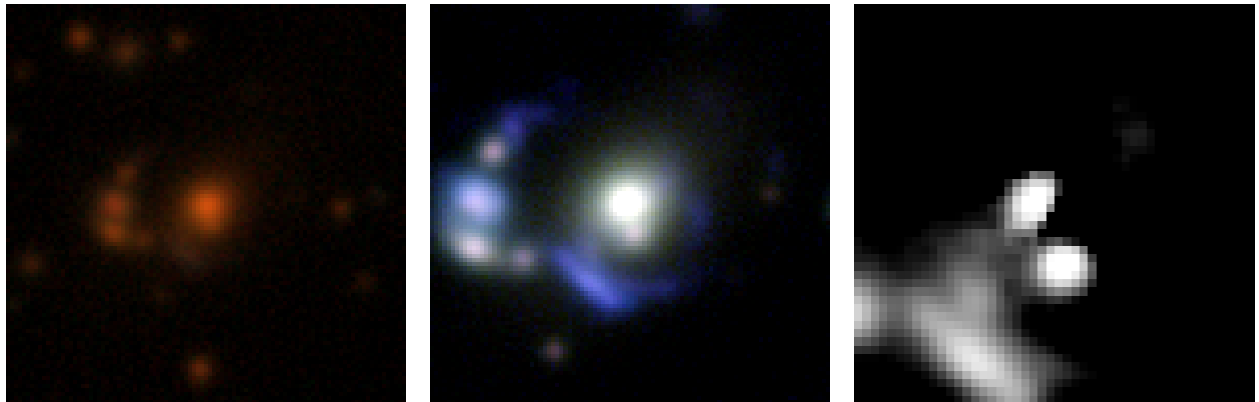


DESI *grz* (0.262''/px)

HSC *grizy* (0.168''/px)

HSC lens-light subtracted

Figure 5: s_351303_461 (RA 194.5345, Dec +3.5358) — HSC tier-2 grade **A**, p_{lens} 0.62→0.92, v3blend8 0.89. *Known*: Huang+2020 DESI-194.5344+03.5358 (grade A), 0.2''.



DESI *grz* (0.262''/px)

HSC *grizy* (0.168''/px)

HSC lens-light subtracted

Figure 6: s_345385_1515 (RA 154.3116, Dec +2.4885) — HSC tier-2 grade **A**, p_{lens} 0.10→0.92, v3blend8 0.81. *Known*: Huang+2020 DESI-154.3116+02.4885 (grade C), 0.1''. (= 8th detection s_345385_1514, the adjacent $\sim 1''$ pair, omitted)

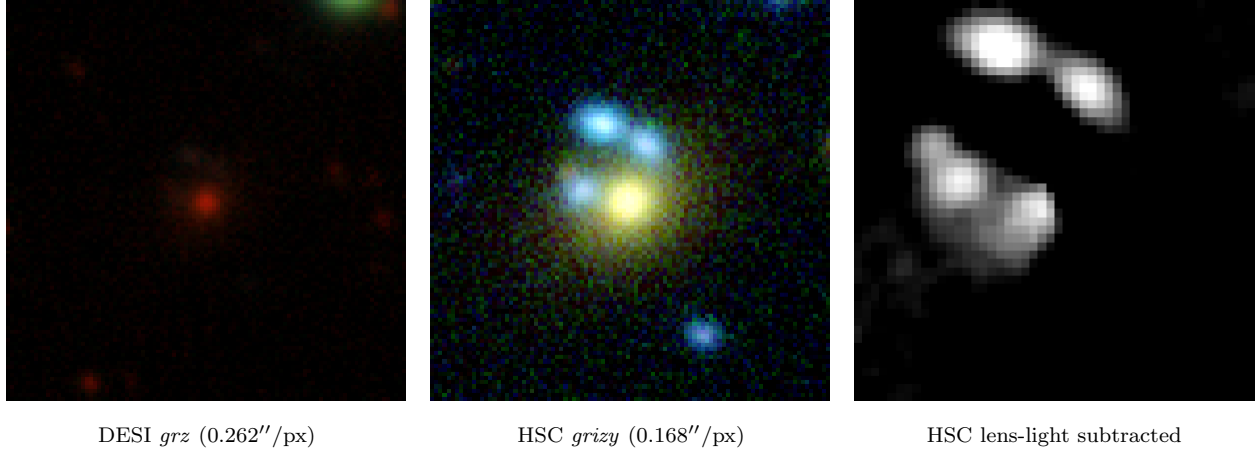


Figure 7: *s*_{340971_2860} (RA 130.8571, Dec +1.7840) — HSC tier-2 grade **A**, p_{lens} 0.12→0.91, *v3blend8* 0.76. *New*: absent from the lineage and from NED/SIMBAD.

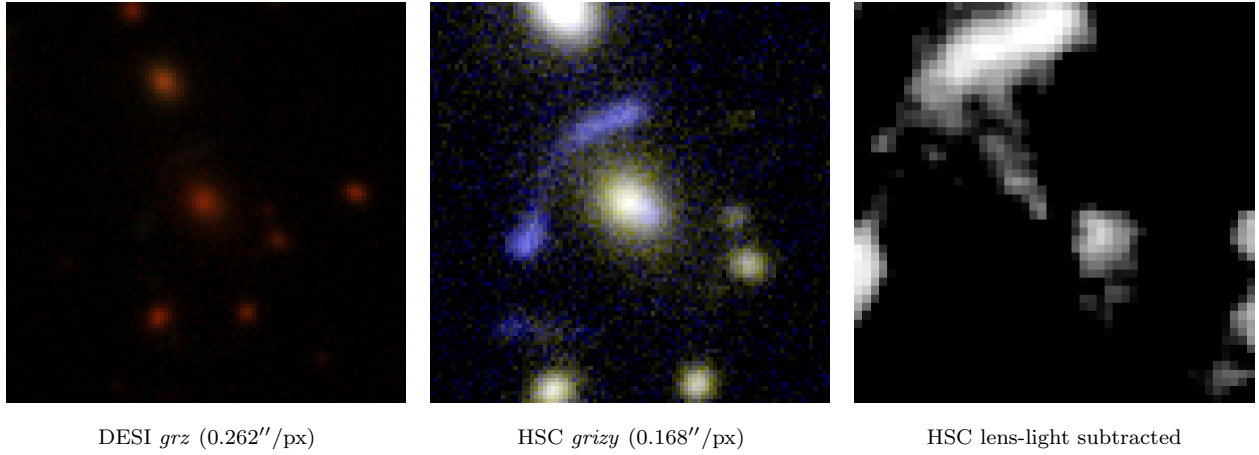


Figure 8: *s*_{326089_1785} (RA 10.2876, Dec -0.7303) — HSC tier-2 grade **A**, p_{lens} 0.03→0.82, *v3blend8* 0.83. *Known*: Huang+2020 DESI-010.2876-00.7303 (grade B), also SOGRAS J0041-0043, 0.1''.

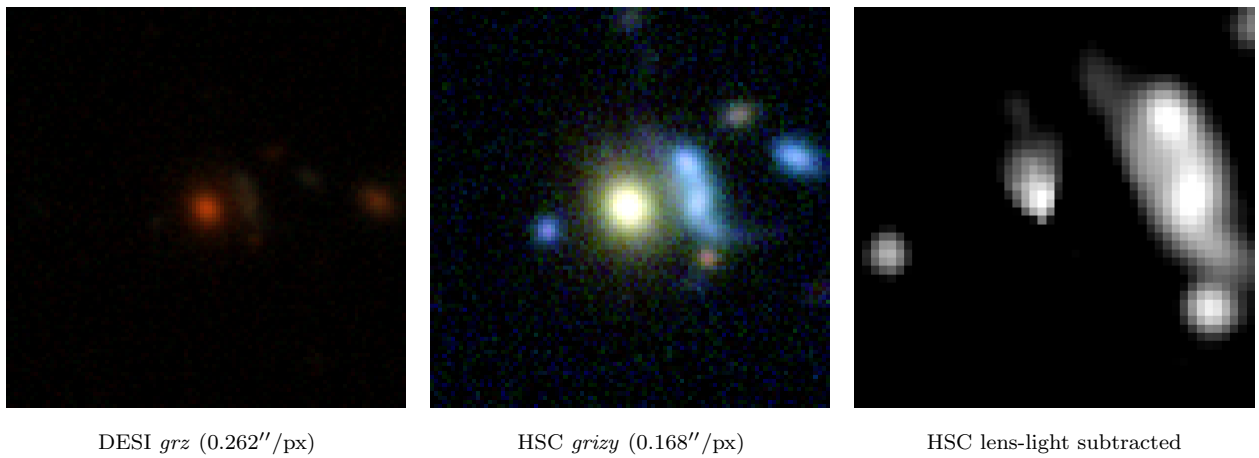


Figure 9: *s*_{325163_2981} (RA 138.8482, Dec -0.9226) — HSC tier-2 grade **A**, p_{lens} 0.18→0.82, *v3blend8* 0.74. *New*: absent from the lineage and from NED/SIMBAD.

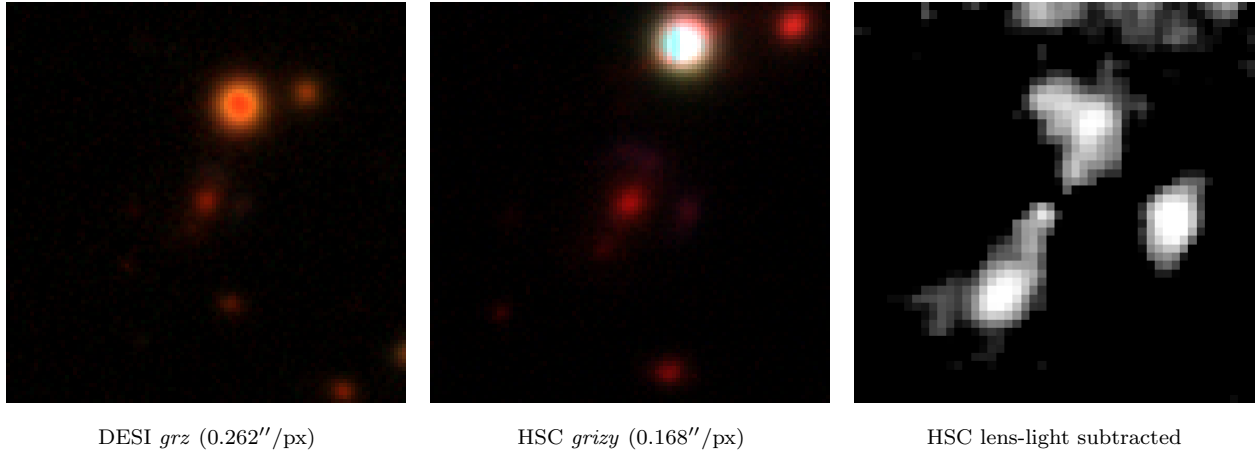


Figure 10: *s_318043_1037* (RA 158.7893, Dec -2.3037) — HSC tier-2 grade **B**, p_{lens} 0.04→0.45, *v3blend8* 0.86. *Known:* Huang+2020 DESI-158.7893-02.3037 (grade B), 0.1''.

B DESI-resolution candidates without high-resolution coverage

Only $\sim 7\%$ of the DR11-south candidates fall in HSC-SSP (§3); the rest cannot be validated beyond DECaLS resolution with current high-res data. Table 4 lists all **78** candidates graded A/B at DECaLS $1''$ by the LensJudge cascade that are *not* in HSC-SSP or Euclid Q1 (8 A, 70 B), ranked by tier-1 p_{lens} . Each is then shown below as an expanded DESI *grz* triptych — *full* (the whole 101 px, $0.262''/\text{px}$ cutout the finder scored), *zoom* ($2.5\times$ center crop), and *residual* (the signed $\chi = (\text{data} - \text{model})/\sigma$ of the $r+z$ luminance after subtracting a robust elliptical-galaxy model, on a red–blue diverging scale at fixed $\pm 5\sigma$: red marks unmodelled excess such as an arc/ring, blue marks over-subtraction; the noise σ is estimated per band, so the residual is honest about significance rather than re-floored to hide its sign) — all at DECaLS resolution, carrying its full metadata and a KNOWN/NEW tag (KNOWN naming the previously-published lens it recovers). The panels are grouped into a KNOWN sub-section (previously-published lenses recovered) followed by a NEW sub-section (candidates new to the literature), ranked by tier-1 p_{lens} within each. A literature crosscheck (`crosscheck_candidates.py`; Huang/Storfer/Inchausti + NED/SIMBAD) finds **53 of the 78 are previously-published lenses** — 38 from the Dark Energy Survey lens searches, 8 Huang+2020, 6 AGEL, 1 CASSOWARY (the *known* column of Table 4) — because this DECaLS-south footprint heavily overlaps DES, which the campaign crossmatch omitted; the **25 genuinely new** are all grade B (the 8 grade-A are all known lenses). A follow-up NED cone search corroborates the new set (no NED lens match; 24/25 returned, one NED timeout). At $1''$ a genuine lens and an lrg+companion mimic are not separable (the resolution ceiling of §3), so these are *candidates*, an unknown and likely large fraction of which are mimics — the per-candidate panels show both clear arc/ring morphologies and ambiguous lrg+companion configurations. They are listed in full for completeness and as a follow-up target set, awaiting Euclid DR1 / Rubin / Roman or targeted high-resolution and spectroscopic follow-up.

Table 4: The 78 DESI-resolution grade-A/B candidates (LensJudge `direct/escalate` grade at DECaLS $0.262''$) with *no* HSC-SSP or Euclid Q1 coverage, ranked by tier-1 p_{lens} . A literature crosscheck (`crosscheck_candidates.py`: the Huang/Storfer/Inchausti catalogues + a NED/SIMBAD cone search) finds **53 are previously-published lenses** — 38 DES, 8 Huang+2020, 6 AGEL, 1 CASSOWARY (*known* column: survey code; per-object matches in `data/dr11s_resolution_xmatch.csv`) — and **25 are genuinely new** (—), all grade B. The high known fraction is because this DECaLS-south footprint heavily overlaps the Dark Energy Survey lens searches, which the campaign crossmatch omitted. The 25 new ones cannot be validated beyond DECaLS resolution; at $1''$ a genuine lens and an lrg+companion mimic are not separable, so an unknown fraction are mimics. Per-candidate full/zoom/residual panels follow (App. B).

#	name	RA	Dec	grade	p_{lens}	v3blend8	known
1	s_310364_6388	38.2078	-3.3906	A	0.90	0.82	H20
2	s_137502_1047	26.4450	-35.6910	A	0.90	0.80	DES
3	s_181942_4122	61.6016	-26.7733	A	0.87	0.86	DES
4	s_441355_1188	177.1381	+19.5009	A	0.87	0.83	CSWA
5	s_98012_247	56.1462	-44.7956	A	0.84	0.86	DES
6	s_124958_7349	58.1768	-38.4292	A	0.81	0.89	DES
7	s_191004_9637	56.9356	-24.9088	A	0.80	0.88	DES
8	s_80200_6872	15.4919	-49.2940	A	0.78	0.88	DES
9	s_230539_4159	222.0715	-17.8489	B	0.72	0.84	—
10	s_180593_5604	44.0974	-27.1218	B	0.63	0.88	DES

continued...

Table 4 continued

#	name	RA	Dec	grade	p_{lens}	v3blend8	known
11	s_292418_3907	216.8280	-6.7541	B	0.62	0.89	H20
12	s_232538_8716	25.2755	-17.2232	B	0.62	0.85	H20
13	s_234689_542	227.0427	-16.8771	B	0.62	0.80	—
14	s_42996_3467	53.7444	-60.5805	B	0.60	0.90	DES
15	s_205319_6061	359.9772	-22.4596	B	0.60	0.88	—
16	s_188493_9685	83.4556	-25.6151	B	0.60	0.88	DES
17	s_38835_314	56.6543	-61.9792	B	0.60	0.84	DES
18	s_218895_5088	32.7525	-19.6362	B	0.60	0.81	—
19	s_240846_8850	30.2832	-15.8547	B	0.60	0.80	H20
20	s_111542_5062	4.8181	-41.6141	B	0.60	0.77	DES
21	s_290988_3355	217.1429	-7.0963	B	0.58	0.75	H20
22	s_194300_2796	242.6052	-24.4155	B	0.58	0.74	—
23	s_50907_11900	303.5808	-57.9504	B	0.58	0.74	DES
24	s_34114_11363	353.7468	-64.0687	B	0.55	0.90	DES
25	s_61626_1035	65.1782	-54.3770	B	0.55	0.87	DES
26	s_159172_8702	37.5699	-31.3669	B	0.55	0.82	DES
27	s_197474_9756	30.7668	-23.6341	B	0.55	0.81	DES
28	s_23764_4161	132.6603	-68.1876	B	0.55	0.79	—
29	s_310225_1602	3.2923	-3.5960	B	0.52	0.75	DES
30	s_61472_9399	359.5200	-54.6765	B	0.50	0.88	DES
31	s_137518_1133	31.3588	-35.6632	B	0.50	0.84	DES
32	s_353585_141	45.8222	+3.9613	B	0.50	0.84	—
33	s_43736_6056	67.3945	-60.1905	B	0.50	0.83	DES
34	s_257579_1725	23.3422	-12.8670	B	0.50	0.82	DES
35	s_289350_3716	164.8051	-7.1449	B	0.50	0.81	—
36	s_148704_6732	183.8176	-33.6006	B	0.50	0.80	—
37	s_301694_7426	26.2679	-4.9310	B	0.48	0.85	H20
38	s_58263_3751	42.7179	-55.4032	B	0.45	0.88	DES
39	s_194944_4494	59.2045	-24.1447	B	0.45	0.85	DES
40	s_280236_3353	25.8623	-8.8393	B	0.45	0.84	AGEL
41	s_90098_8002	107.1696	-46.7832	B	0.45	0.83	—
42	s_82090_1248	14.4037	-48.8040	B	0.45	0.81	DES
43	s_502450_1680	161.1145	+31.2344	B	0.45	0.79	—
44	s_426295_4804	182.0229	+16.6326	B	0.45	0.79	—
45	s_369309_653	30.4465	+6.6551	B	0.45	0.74	—
46	s_32666_26056	248.6090	-64.5569	B	0.45	0.73	—
47	s_156771_3446	54.3219	-31.8704	B	0.42	0.89	AGEL
48	s_94861_9433	21.9716	-45.5428	B	0.42	0.87	DES
49	s_102952_10164	334.8017	-43.8098	B	0.42	0.85	DES
50	s_90804_9850	3.9284	-46.6031	B	0.42	0.84	DES
51	s_71042_1653	20.1760	-51.7314	B	0.42	0.82	DES
52	s_266620_918	172.1320	-11.3108	B	0.42	0.80	—
53	s_242312_4967	50.3567	-15.5010	B	0.40	0.90	—
54	s_381188_1199	144.1511	+8.8633	B	0.40	0.90	H20

continued...

Table 4 continued

#	name	RA	Dec	grade	p_{lens}	v3blend8	known
55	s_235296_3732	25.6457	-16.8049	B	0.40	0.88	AGEL
56	s_262449_2840	188.7861	-11.9798	B	0.40	0.87	—
57	s_91761_4679	350.3683	-46.5138	B	0.40	0.86	DES
58	s_360639_3636	13.4868	+5.3590	B	0.40	0.85	—
59	s_107312_5223	23.8451	-42.5398	B	0.40	0.85	DES
60	s_69087_268	307.2325	-52.5218	B	0.40	0.81	DES
61	s_165468_9648	64.7771	-29.9937	B	0.40	0.78	DES
62	s_256199_9097	29.6819	-13.0258	B	0.40	0.78	DES
63	s_364017_5199	141.0626	+5.7690	B	0.40	0.77	H20
64	s_188312_3322	33.2537	-25.3945	B	0.40	0.76	—
65	s_188754_2468	155.4695	-25.5029	B	0.40	0.74	—
66	s_119342_4136	50.7164	-39.7694	B	0.38	0.90	DES
67	s_405220_1292	136.4655	+13.0307	B	0.38	0.89	—
68	s_209576_5226	65.5760	-21.5461	B	0.38	0.86	DES
69	s_225679_7025	25.7203	-18.5211	B	0.38	0.79	AGEL
70	s_250611_2161	33.7898	-14.1185	B	0.38	0.77	—
71	s_161725_5181	61.4085	-30.6887	B	0.38	0.76	—
72	s_210799_616	33.4979	-21.2827	B	0.38	0.75	—
73	s_129372_3768	16.4504	-37.4286	B	0.35	0.82	DES
74	s_238185_2253	58.5761	-16.1645	B	0.35	0.82	AGEL
75	s_94933_7964	47.5548	-45.5743	B	0.35	0.78	DES
76	s_250997_4626	133.2708	-13.8944	B	0.35	0.77	—
77	s_71871_246	353.9664	-51.8716	B	0.35	0.76	AGEL
78	s_90920_9419	45.9517	-46.4407	B	0.30	0.85	DES

Known — previously-published lenses recovered (53 of 78)

These coincide ($\leq \text{few}''$) with a catalogued strong lens — 38 DES, 8 Huang+2020, 6 AGEL, 1 CASSOWARY — an external validation of the finder (the campaign crossmatch had omitted these catalogues). Ranked by tier-1 p_{lens} .

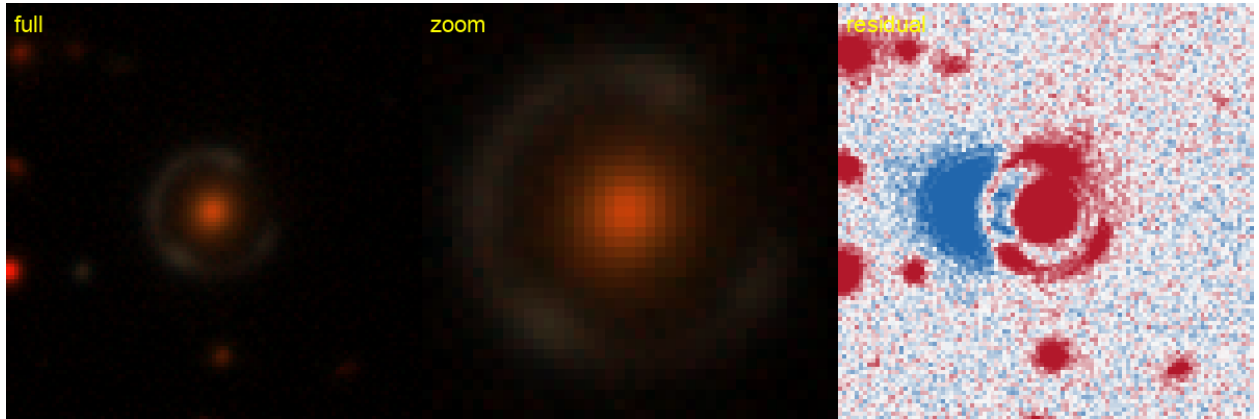


Figure 11: #1 s_310364_6388 — RA 38.2078, Dec -3.3906; grade **A**, tier-1 p_{lens} 0.90, v3blend8 0.82. **KNOWN** — Huang+2020: DESI-038.2078-03.3906, 0.16'' (lineage).

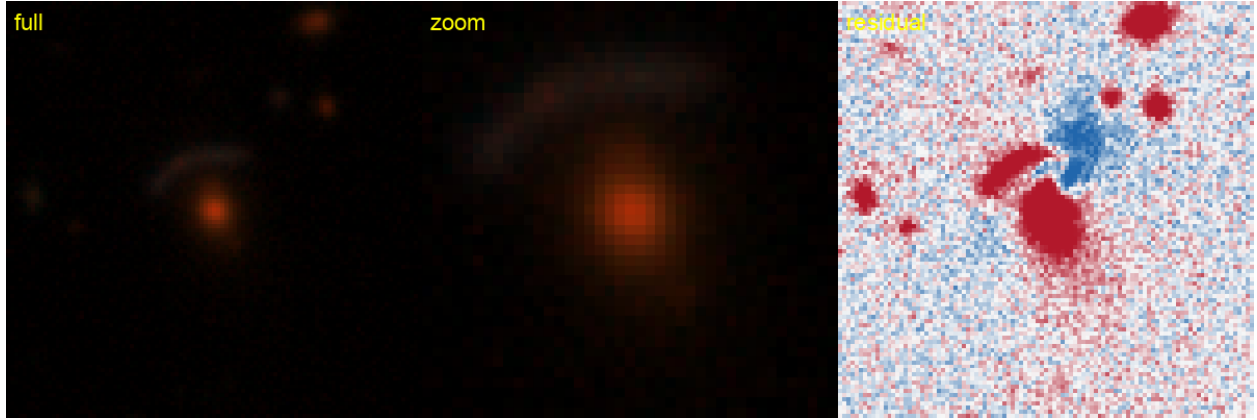


Figure 12: #2 s_137502.1047 — RA 26.4450, Dec -35.6910; grade **A**, tier-1 p_{lens} 0.90, v3blend8 0.80.
KNOWN — DES: DES J014546.78-354127.3, 0.20'' (SIMBAD).

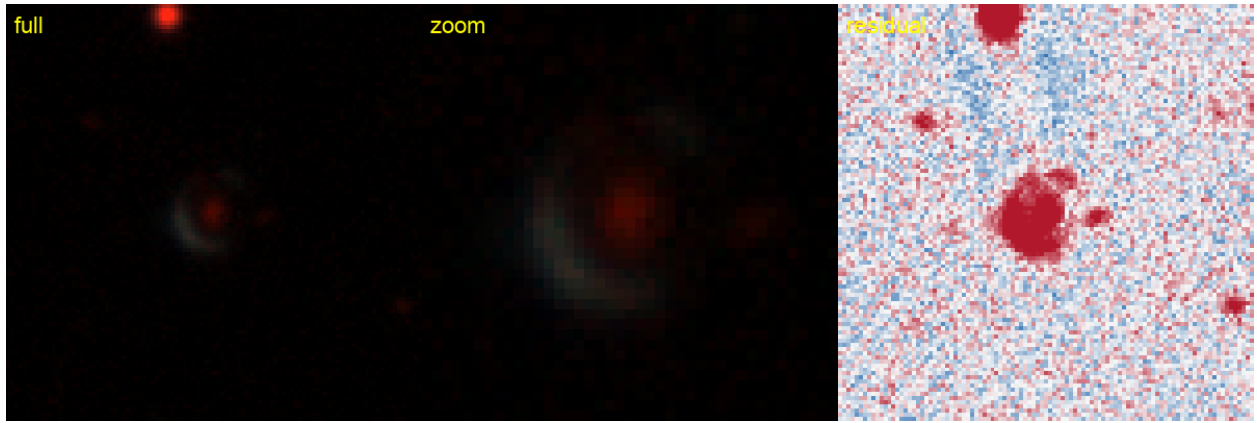


Figure 13: #3 s_181942.4122 — RA 61.6016, Dec -26.7733; grade **A**, tier-1 p_{lens} 0.87, v3blend8 0.86.
KNOWN — DES: DES J040624.49-264625.0, 1.74'' (SIMBAD).

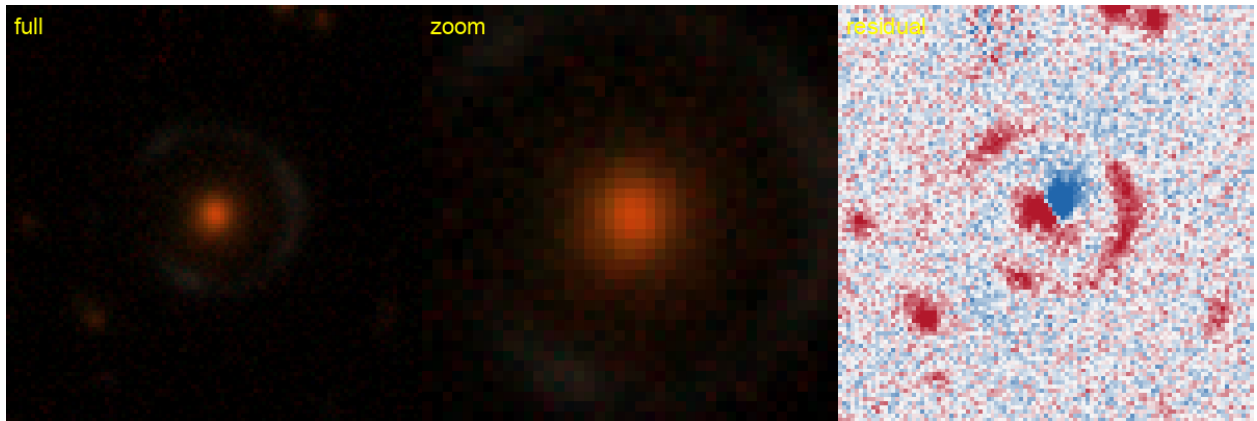


Figure 14: #4 s_441355.1188 — RA 177.1381, Dec +19.5009; grade **A**, tier-1 p_{lens} 0.87, v3blend8 0.83.
KNOWN — CASSOWARY: CSWA 1, 0.01'' (lineage).



Figure 15: #5 s_98012_247 — RA 56.1462, Dec -44.7956; grade **A**, tier-1 p_{lens} 0.84, v3blend8 0.86.
KNOWN — DES: DES J034435.09-444743.9, 0.18'' (SIMBAD).

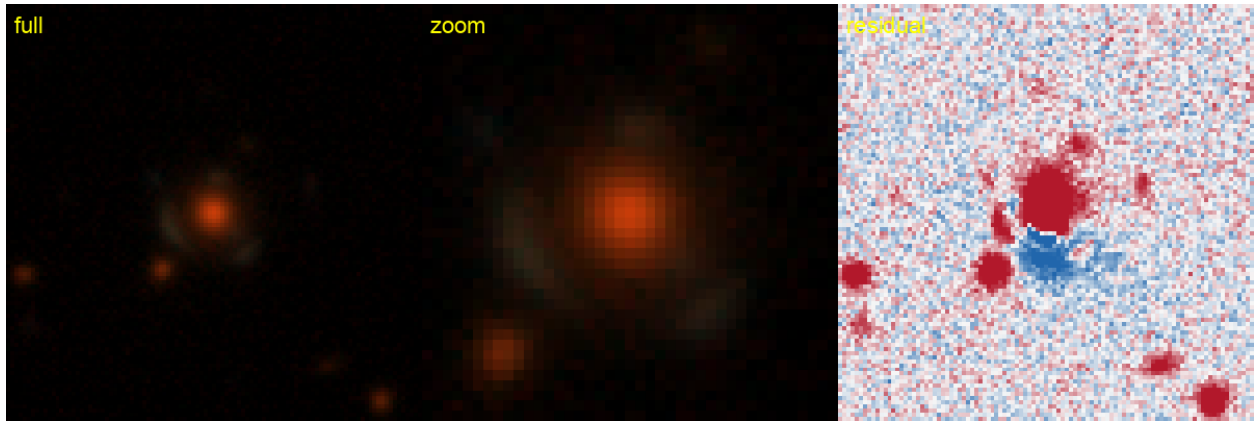


Figure 16: #6 s_124958_7349 — RA 58.1768, Dec -38.4292; grade **A**, tier-1 p_{lens} 0.81, v3blend8 0.89.
KNOWN — DES: DES J035242.40-382544.9, 0.01'' (lineage).

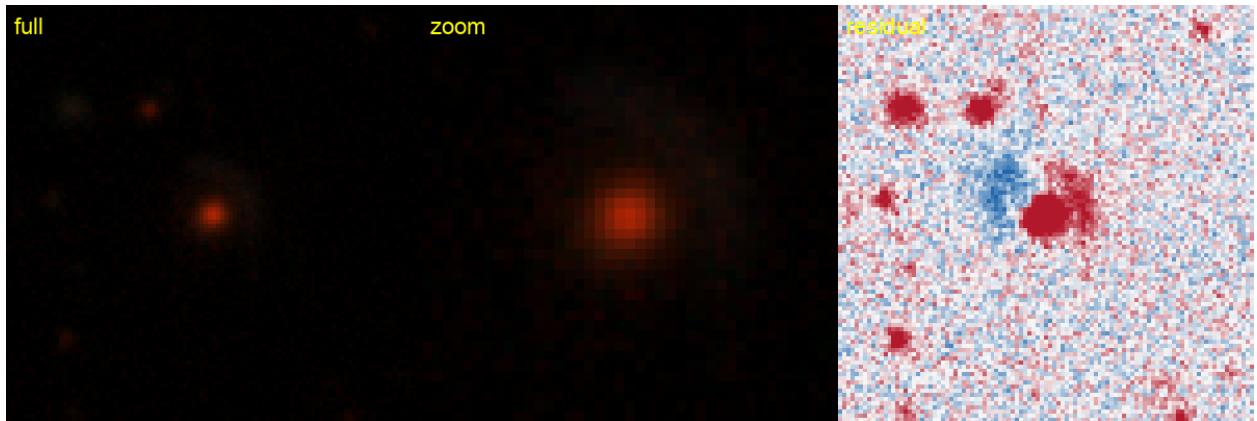


Figure 17: #7 s_191004_9637 — RA 56.9356, Dec -24.9088; grade **A**, tier-1 p_{lens} 0.80, v3blend8 0.88.
KNOWN — DES: DES J034744.53-245431.4, 0.20'' (SIMBAD).

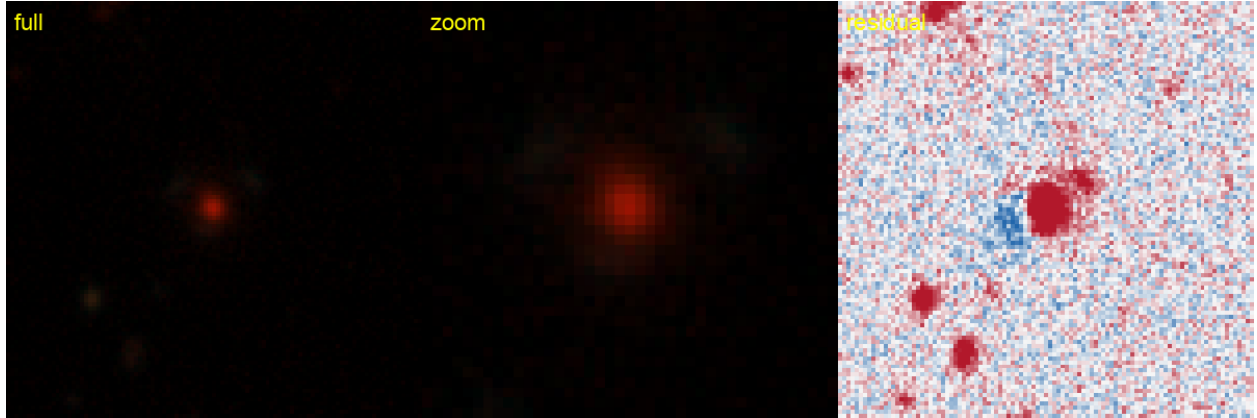


Figure 18: #8 s_80200_6872 — RA 15.4919, Dec -49.2940; grade **A**, tier-1 p_{lens} 0.78, v3blend8 0.88.
KNOWN — DES: DES J010158.03-491738.1, 0.12'' (SIMBAD).

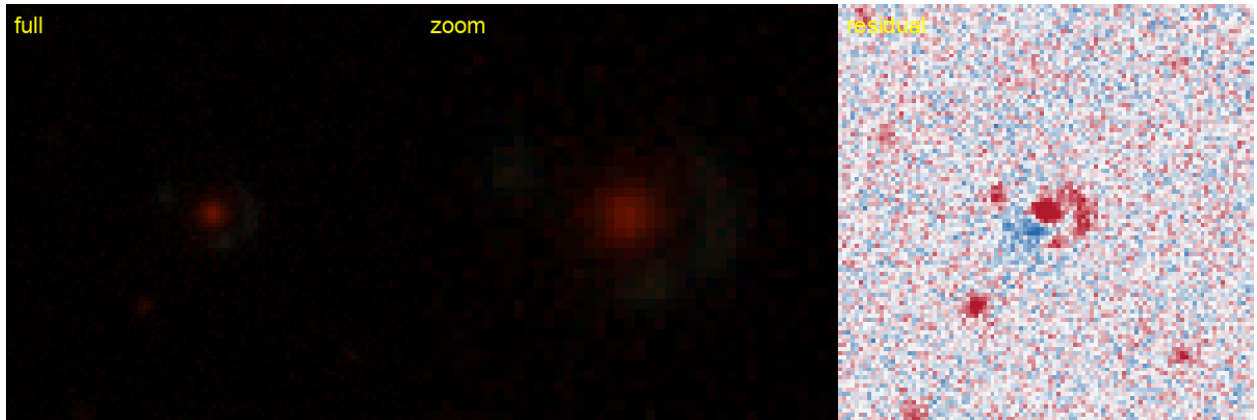


Figure 19: #10 s_180593_5604 — RA 44.0974, Dec -27.1218; grade **B**, tier-1 p_{lens} 0.63, v3blend8 0.88.
KNOWN — DES: DES J025623.36-270718.5, 0.28'' (SIMBAD).

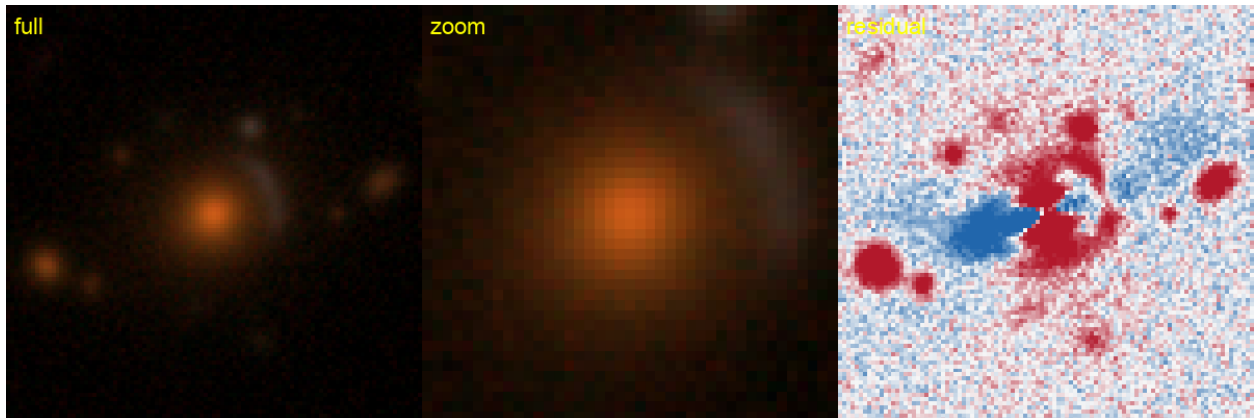


Figure 20: #11 s_292418_3907 — RA 216.8280, Dec -6.7541; grade **B**, tier-1 p_{lens} 0.62, v3blend8 0.89.
KNOWN — Huang+2020: DESI-216.8280-06.7541, 0.15'' (lineage).

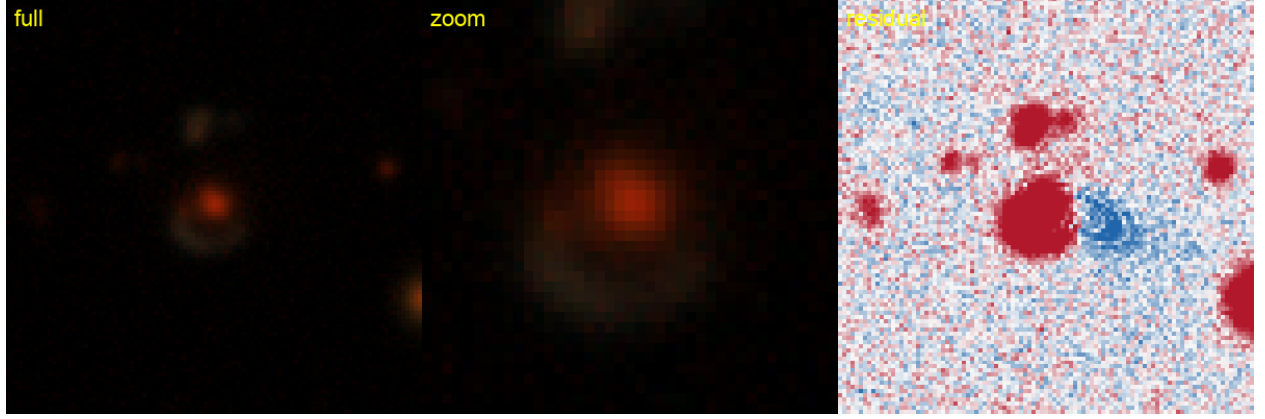


Figure 21: #12 s_232538.8716 — RA 25.2755, Dec -17.2232; grade **B**, tier-1 p_{lens} 0.62, v3blend8 0.85.
KNOWN — Huang+2020: DESI-025.2755-17.2232, 0.19'' (lineage).

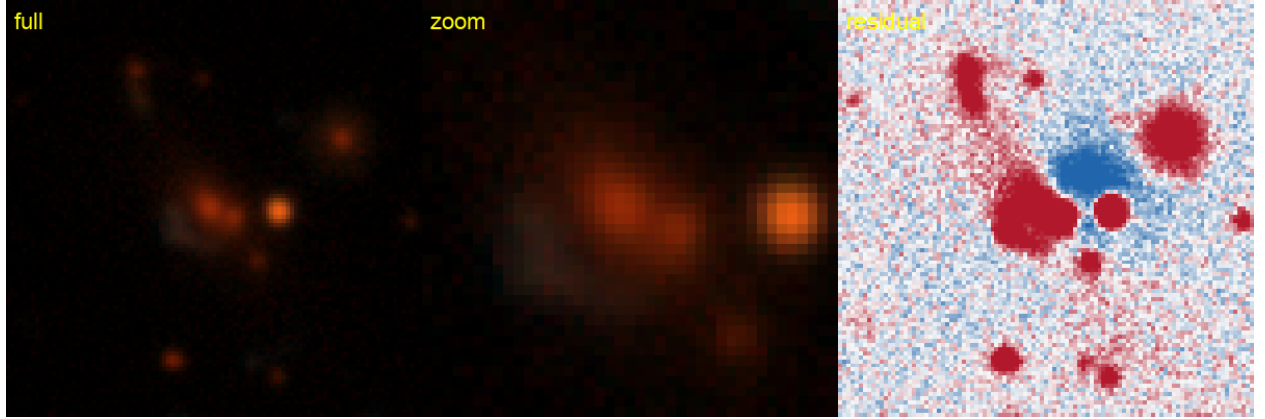


Figure 22: #14 s_42996.3467 — RA 53.7444, Dec -60.5805; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.90.
KNOWN — DES: DES J033458.64-603450.0, 0.15'' (SIMBAD).

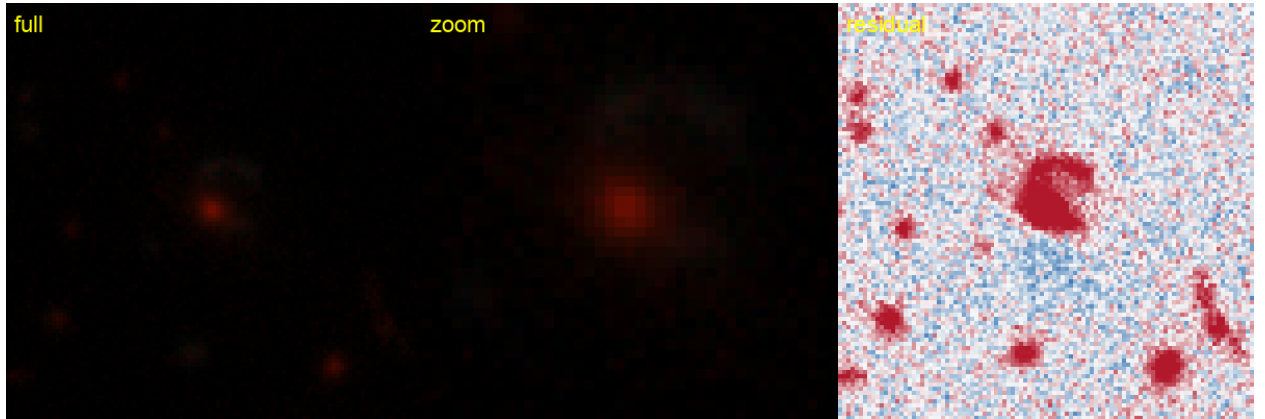


Figure 23: #16 s_188493.9685 — RA 83.4556, Dec -25.6151; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.88.
KNOWN — DES: DES J053349.32-253654.3, 0.15'' (SIMBAD).

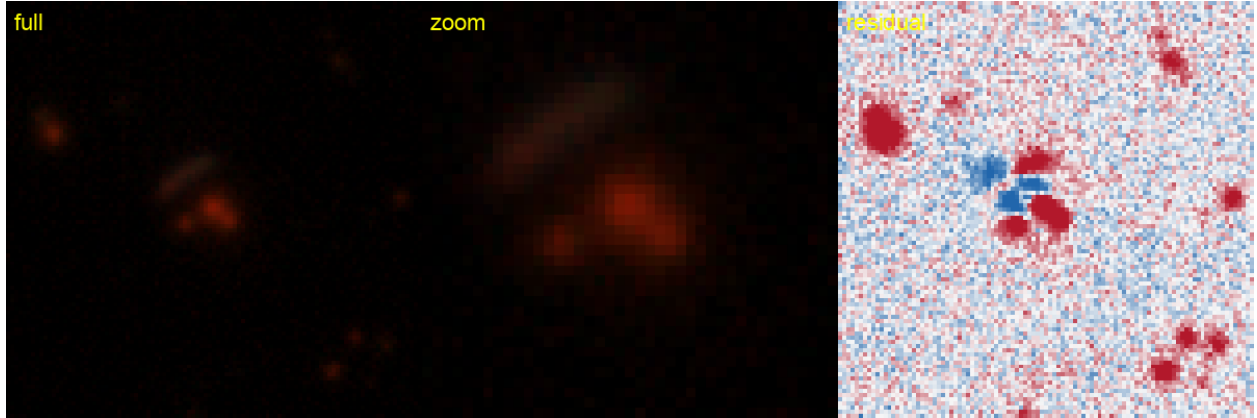


Figure 24: #17 s_38835_314 — RA 56.6543, Dec -61.9792; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.84.
KNOWN — DES: [DBL2017] DES J0346-6158 (A), 0.23'' (SIMBAD).

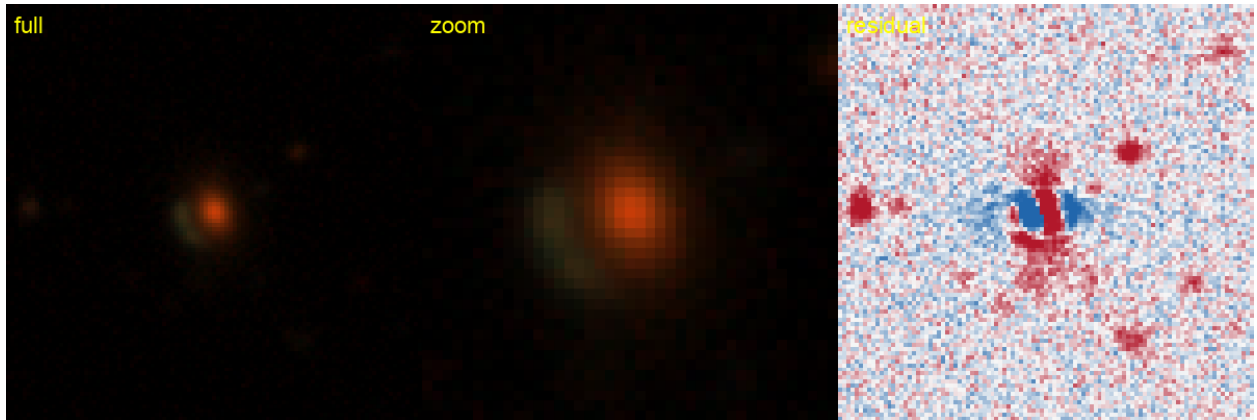


Figure 25: #19 s_240846_8850 — RA 30.2832, Dec -15.8547; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.80.
KNOWN — Huang+2020: DESI-030.2832-15.8547, 0.15'' (lineage).

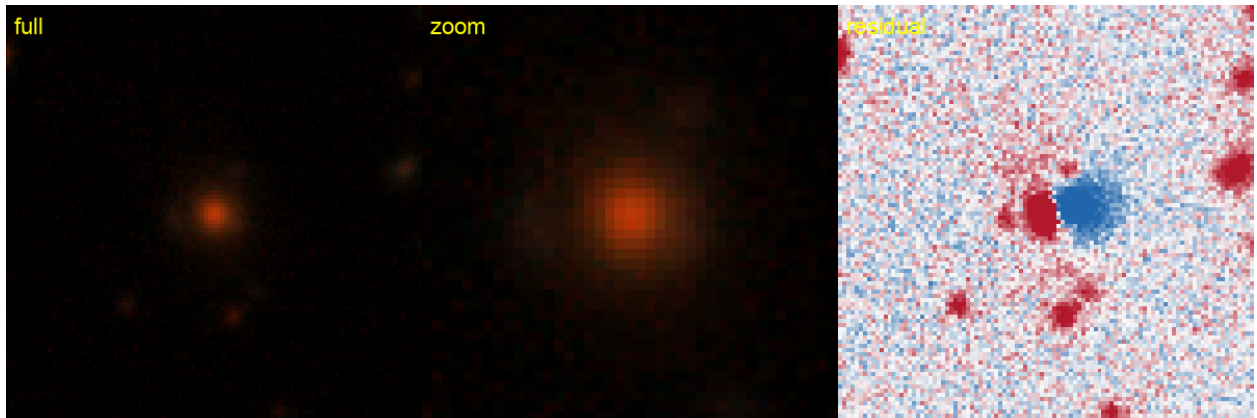


Figure 26: #20 s_111542_5062 — RA 4.8181, Dec -41.6141; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.77.
KNOWN — DES: DES J001916.32-413650.5, 0.22'' (SIMBAD).

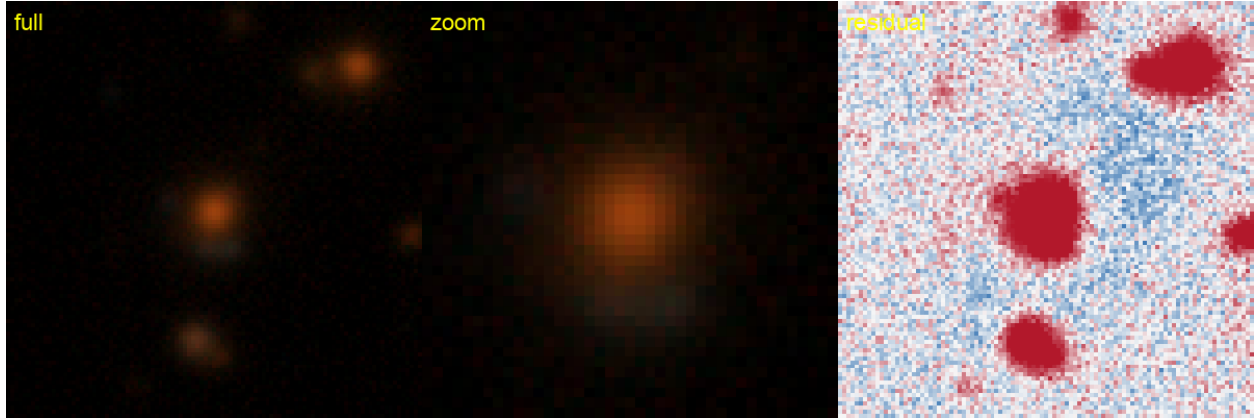


Figure 27: #21 s_290988_3355 — RA 217.1429, Dec -7.0963; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.75.
KNOWN — Huang+2020: DESI-217.1429-07.0963, 0.15'' (lineage).

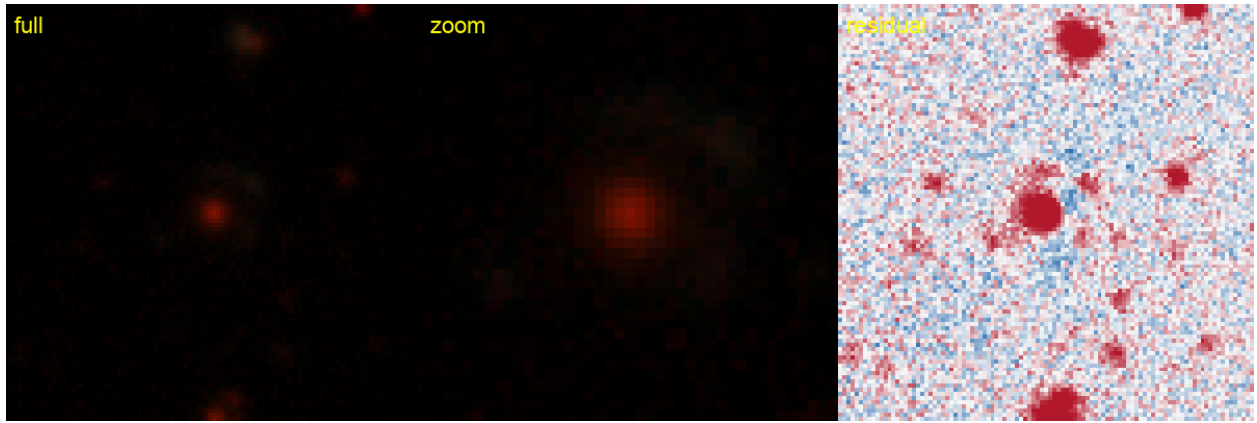


Figure 28: #23 s_50907_11900 — RA 303.5808, Dec -57.9504; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.74.
KNOWN — DES: DES J201419.38-575701.4, 0.15'' (SIMBAD).

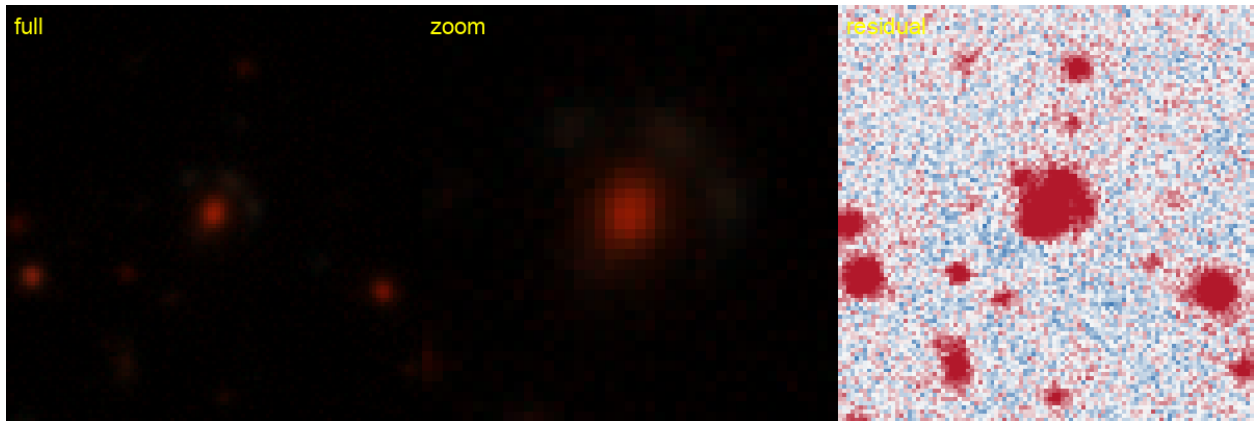


Figure 29: #24 s_34114_11363 — RA 353.7468, Dec -64.0687; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.90.
KNOWN — DES: DES J233459.19-640406.9, 0.32'' (SIMBAD).

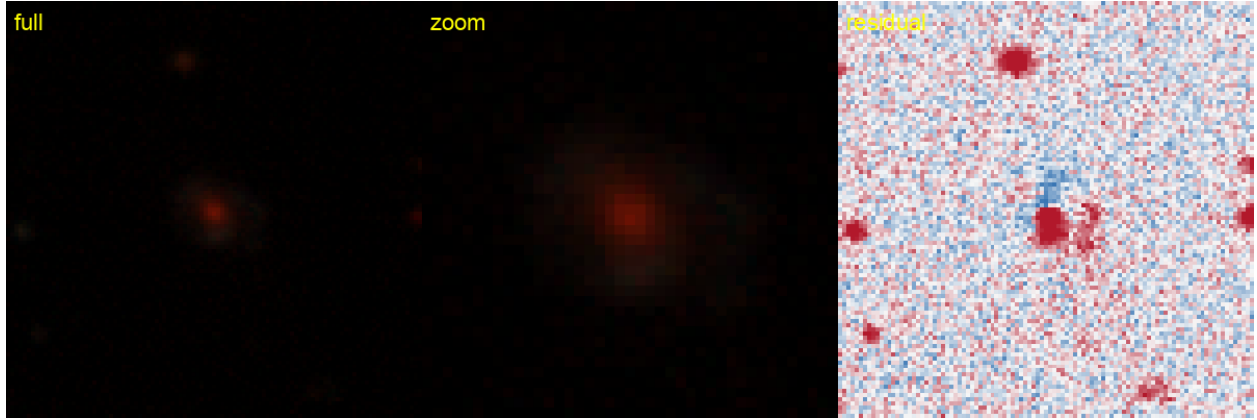


Figure 30: #25 s_61626_1035 — RA 65.1782, Dec -54.3770; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.87.
KNOWN — DES: DES J042042.76-542237.4, 0.18'' (SIMBAD).

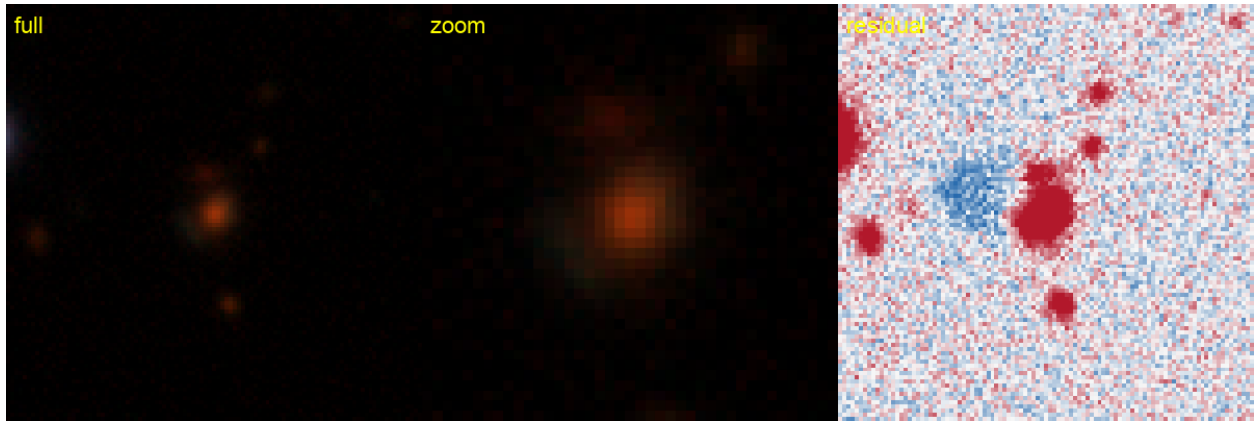


Figure 31: #26 s_159172_8702 — RA 37.5699, Dec -31.3669; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.82.
KNOWN — DES: DES J023016.77-312200.8, 0.07'' (SIMBAD).

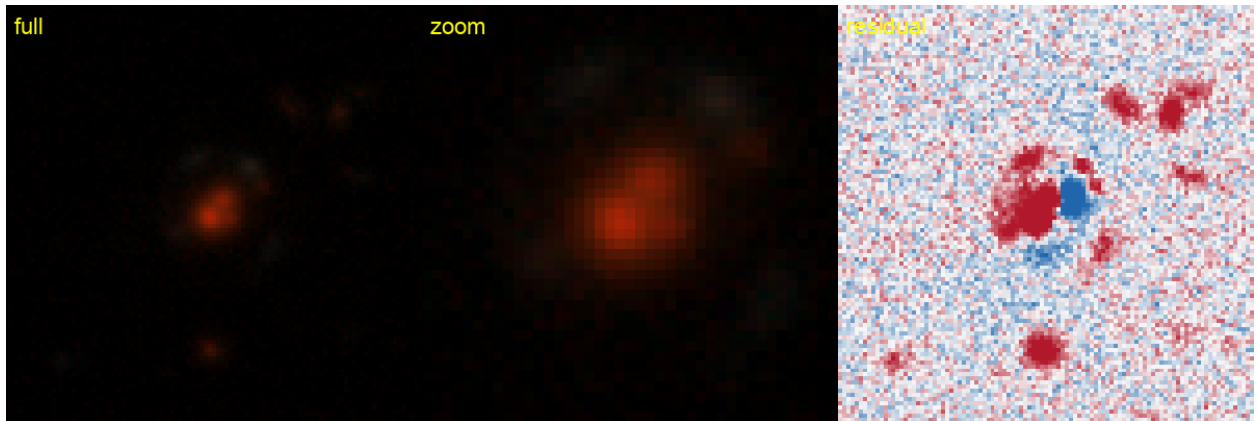


Figure 32: #27 s_197474_9756 — RA 30.7668, Dec -23.6341; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.81.
KNOWN — DES: DES J020304.01-233802.5, 0.22'' (SIMBAD).

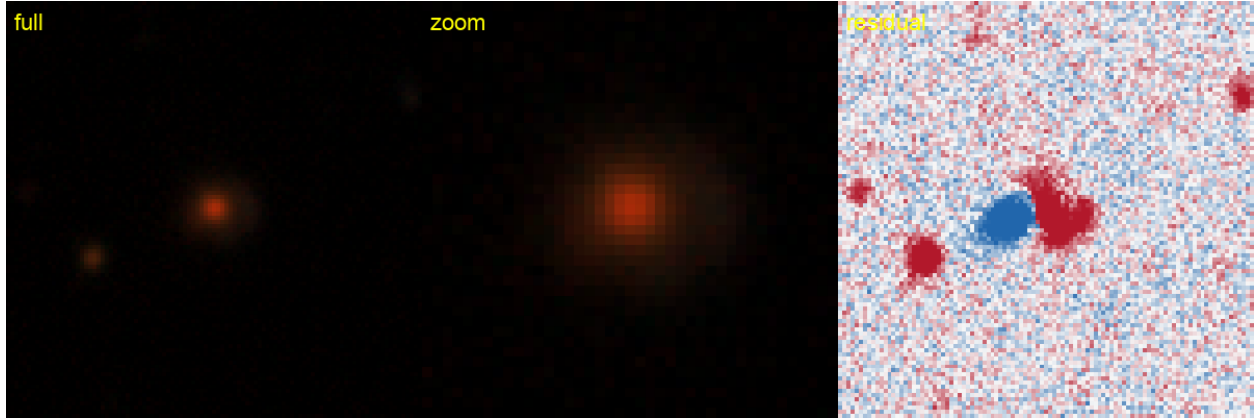


Figure 33: #29 s_310225_1602 — RA 3.2923, Dec -3.5960; grade **B**, tier-1 p_{lens} 0.52, v3blend8 0.75.
KNOWN — DES: DES J001310.13-033545.4, 0.18'' (SIMBAD).

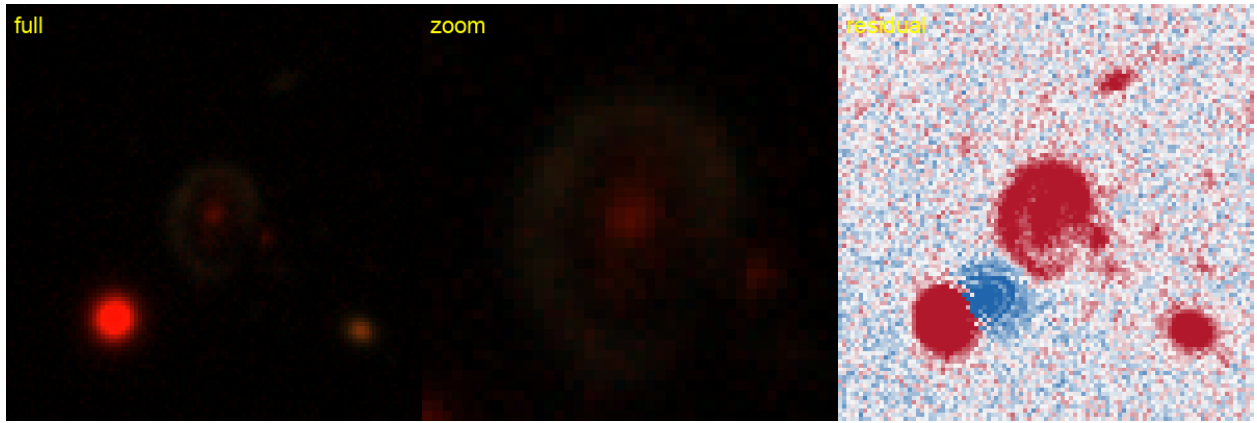


Figure 34: #30 s_61472_9399 — RA 359.5200, Dec -54.6765; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.88.
KNOWN — DES: DES J235804.95-544034.4, 1.73'' (SIMBAD).

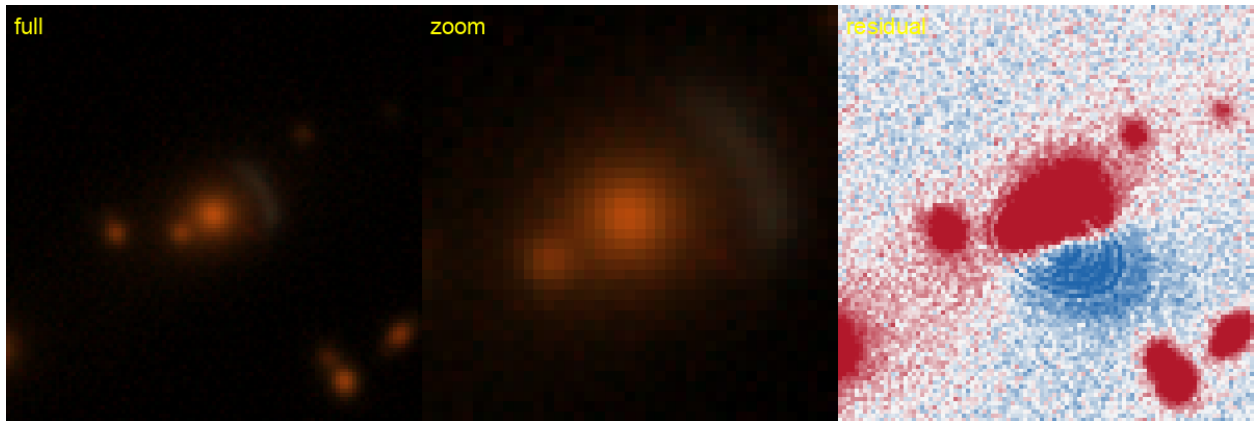


Figure 35: #31 s_137518_1133 — RA 31.3588, Dec -35.6632; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.84.
KNOWN — DES: DES J020526.09-353947.4, 0.14'' (SIMBAD).

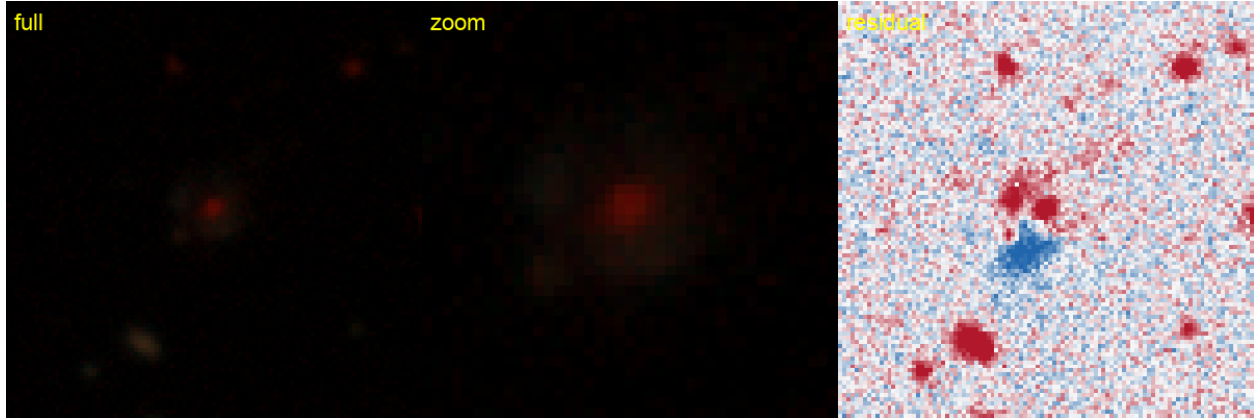


Figure 36: #33 s_43736.6056 — RA 67.3945, Dec -60.1905; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.83.
KNOWN — DES: DES J042934.66-601126.3, 0.35'' (SIMBAD).

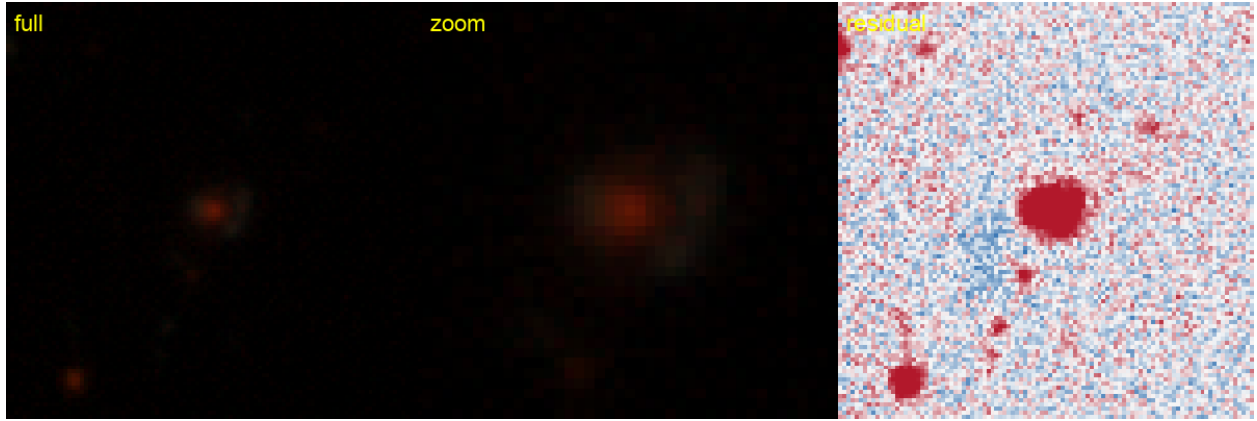


Figure 37: #34 s_257579.1725 — RA 23.3422, Dec -12.8670; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.82.
KNOWN — DES: DES J013322.10-125201.2, 0.27'' (SIMBAD).

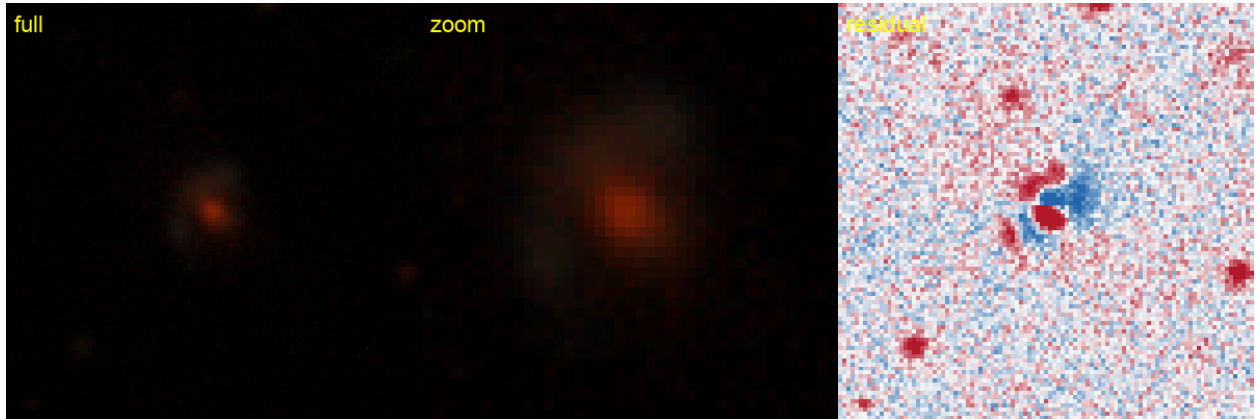


Figure 38: #37 s_301694.7426 — RA 26.2679, Dec -4.9310; grade **B**, tier-1 p_{lens} 0.48, v3blend8 0.85.
KNOWN — Huang+2020: DESI-026.2679-04.9310, 0.13'' (lineage).

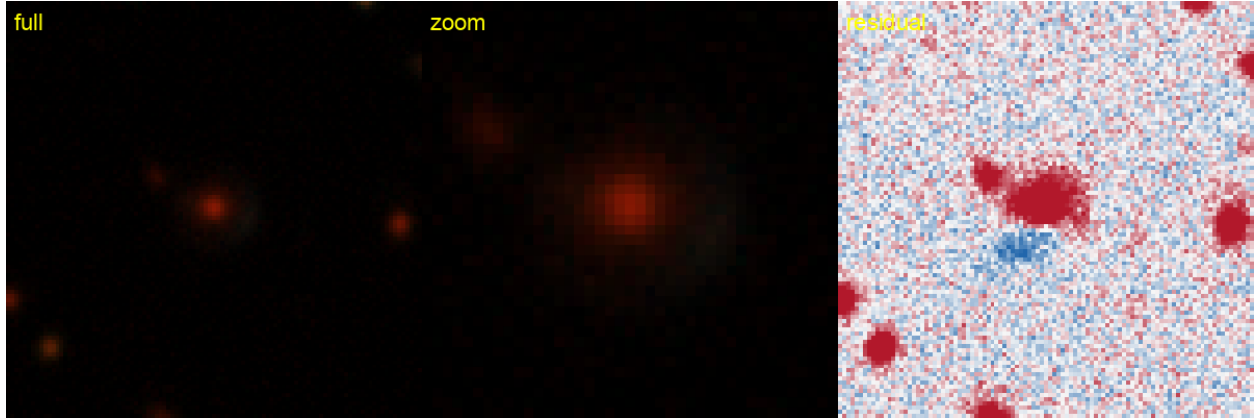


Figure 39: #38 s_58263.3751 — RA 42.7179, Dec -55.4032; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.88.
KNOWN — DES: DES J025052.27-552411.7, 0.22'' (SIMBAD).

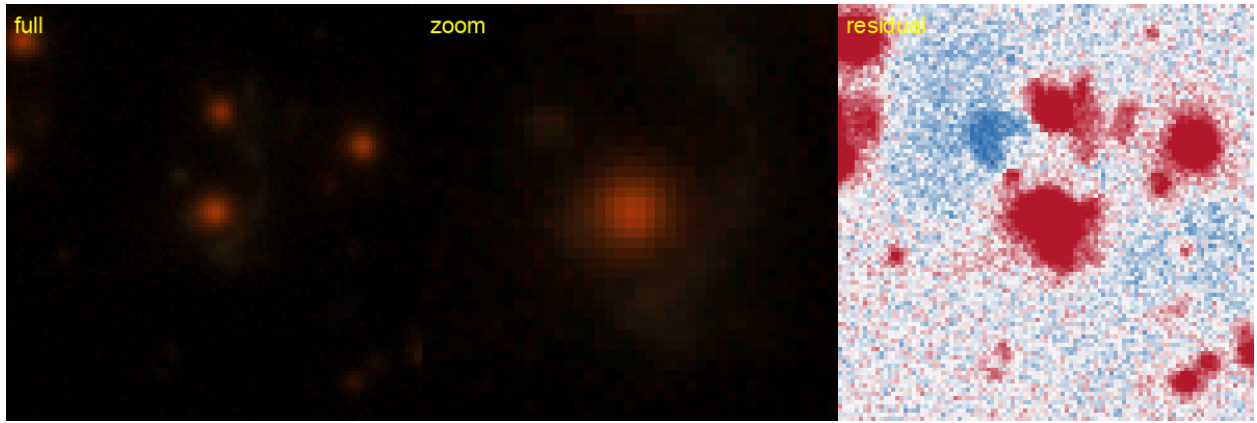


Figure 40: #39 s_194944.4494 — RA 59.2045, Dec -24.1447; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.85.
KNOWN — DES: DES J035649.05-240841.1, 0.28'' (SIMBAD).

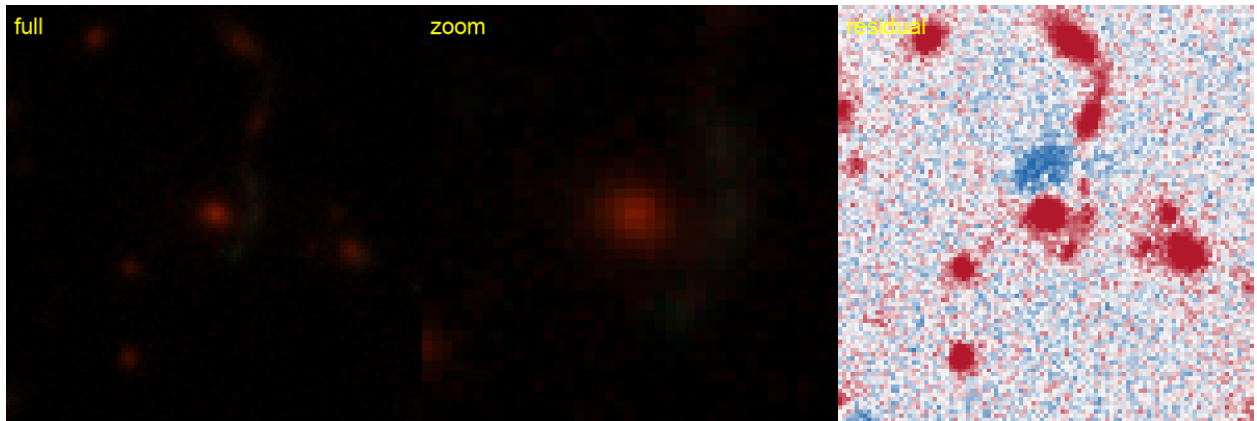


Figure 41: #40 s_280236.3353 — RA 25.8623, Dec -8.8393; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.84.
KNOWN — AGEL: AGEL J014327-085021 system, 0.93'' (SIMBAD).

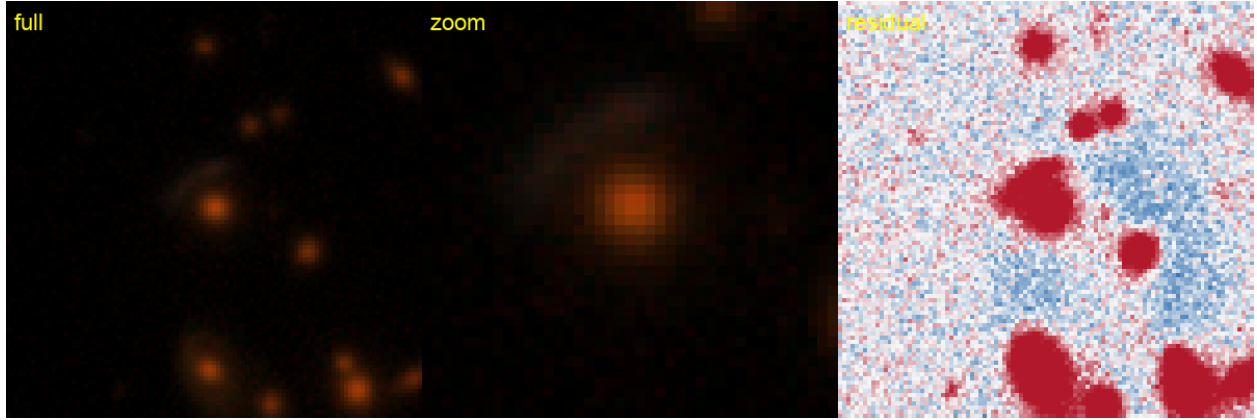


Figure 42: #42 s_82090.1248 — RA 14.4037, Dec -48.8040; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.81.
KNOWN — DES: [DBL2017] DES J0057-4848 (A), 0.19'' (SIMBAD).

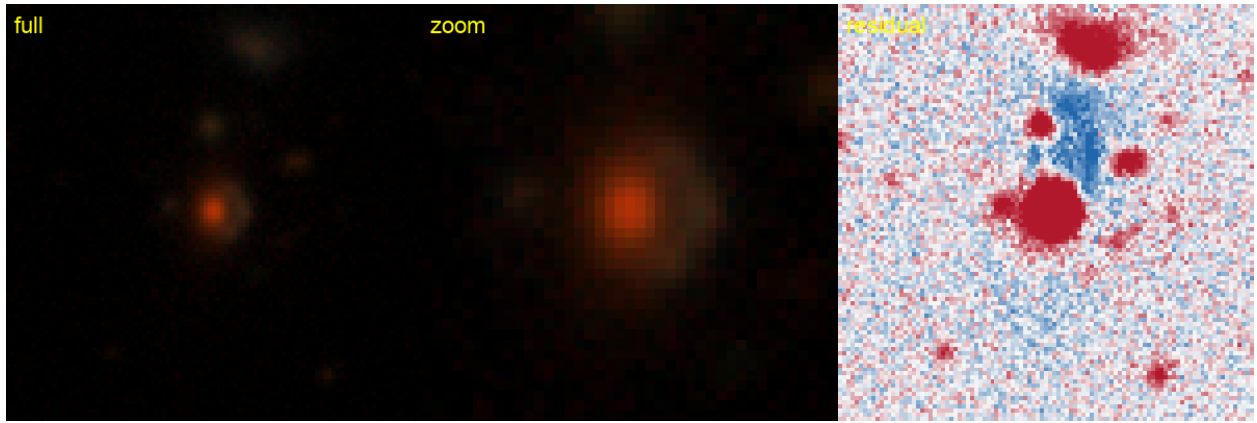


Figure 43: #47 s_156771.3446 — RA 54.3219, Dec -31.8704; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.89.
KNOWN — AGEL: AGEL J033717-315214 system, 0.99'' (SIMBAD).

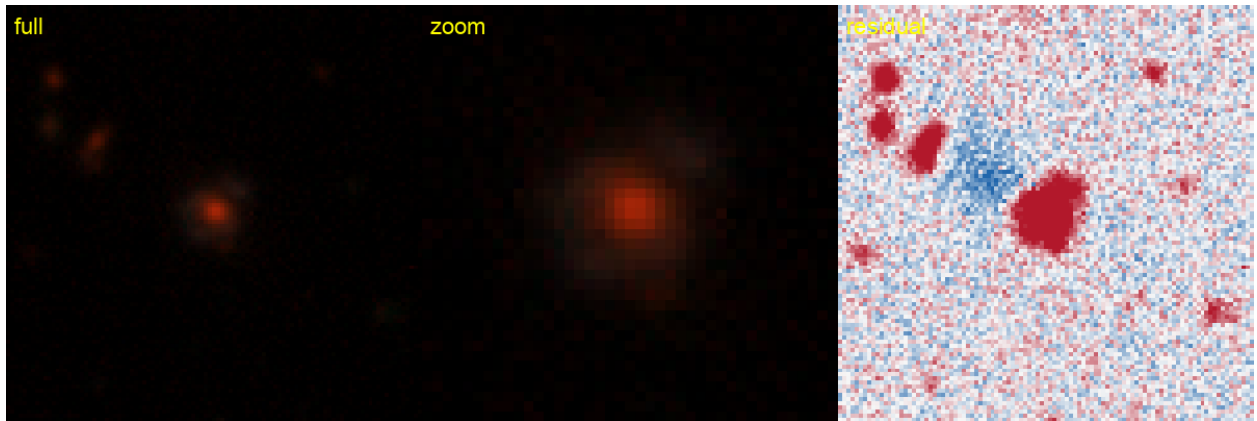


Figure 44: #48 s_94861.9433 — RA 21.9716, Dec -45.5428; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.87.
KNOWN — DES: DES J012753.17-453233.8, 0.17'' (SIMBAD).

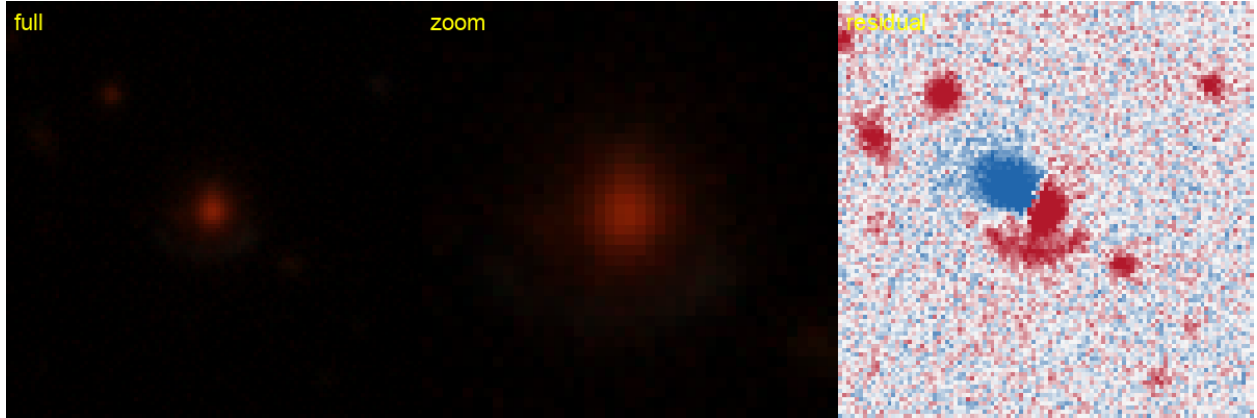


Figure 45: #49 s_102952_10164 — RA 334.8017, Dec -43.8098; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.85.
KNOWN — DES: DES J221912.39-434835.1, 0.01'' (lineage).

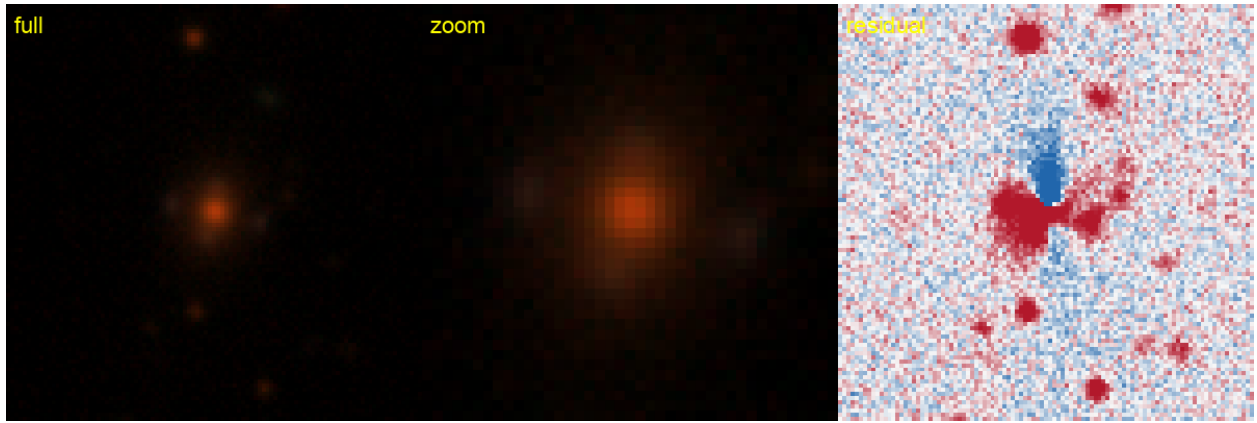


Figure 46: #50 s_90804_9850 — RA 3.9284, Dec -46.6031; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.84.
KNOWN — DES: DES J001542.79-463610.9, 0.16'' (SIMBAD).

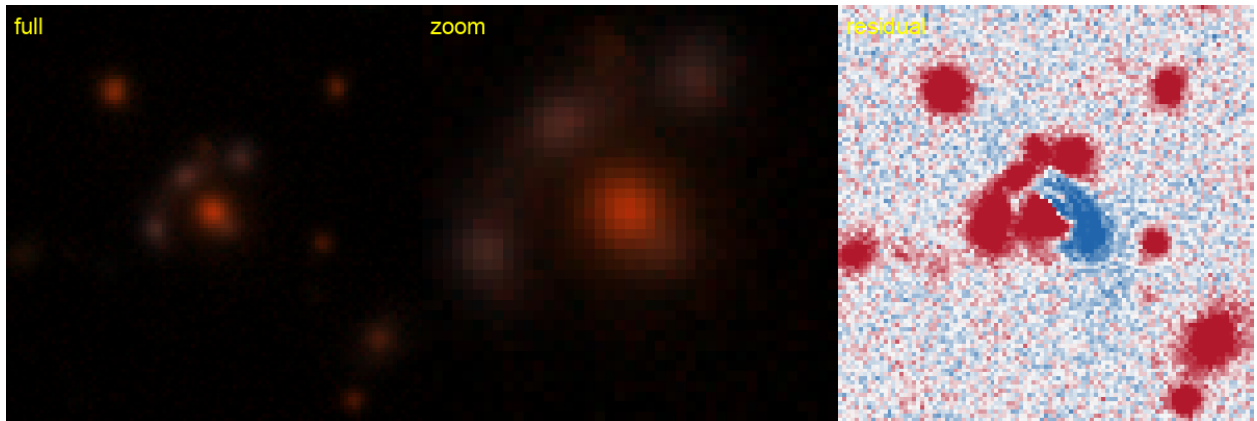


Figure 47: #51 s_71042_1653 — RA 20.1760, Dec -51.7314; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.82.
KNOWN — DES: [DBL2017] DES J0120-5143 (A), 0.19'' (SIMBAD).

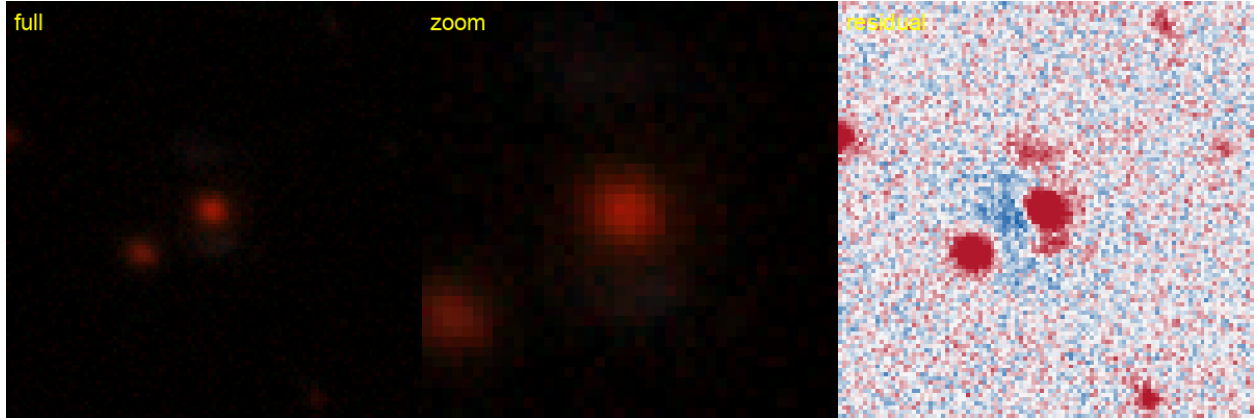


Figure 48: #54 s_381188_1199 — RA 144.1511, Dec +8.8633; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.90.
KNOWN — Huang+2020: DESI-144.1511+08.8633, 0.20'' (lineage).

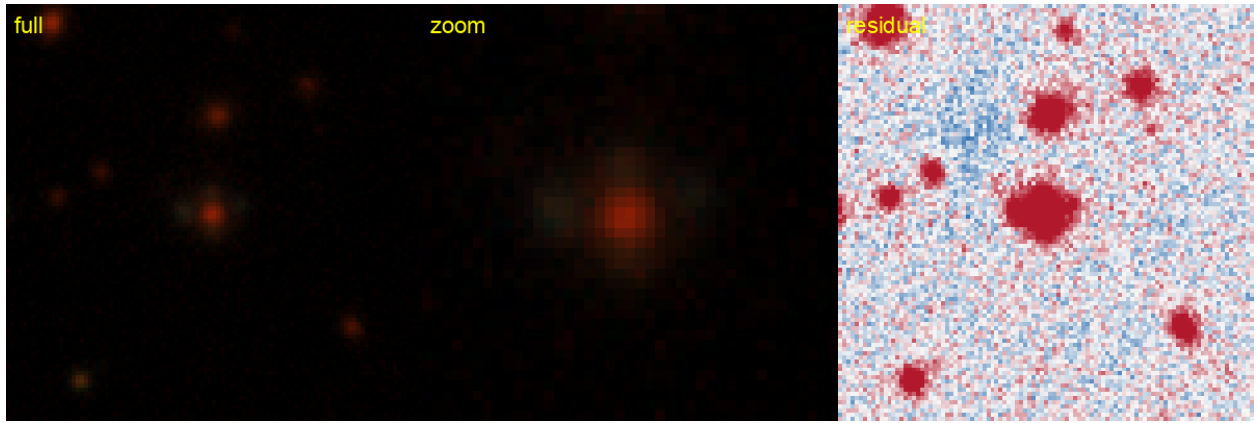


Figure 49: #55 s_235296_3732 — RA 25.6457, Dec -16.8049; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.88.
KNOWN — AGEL: AGEL J014235-164818 system, 0.86'' (SIMBAD).

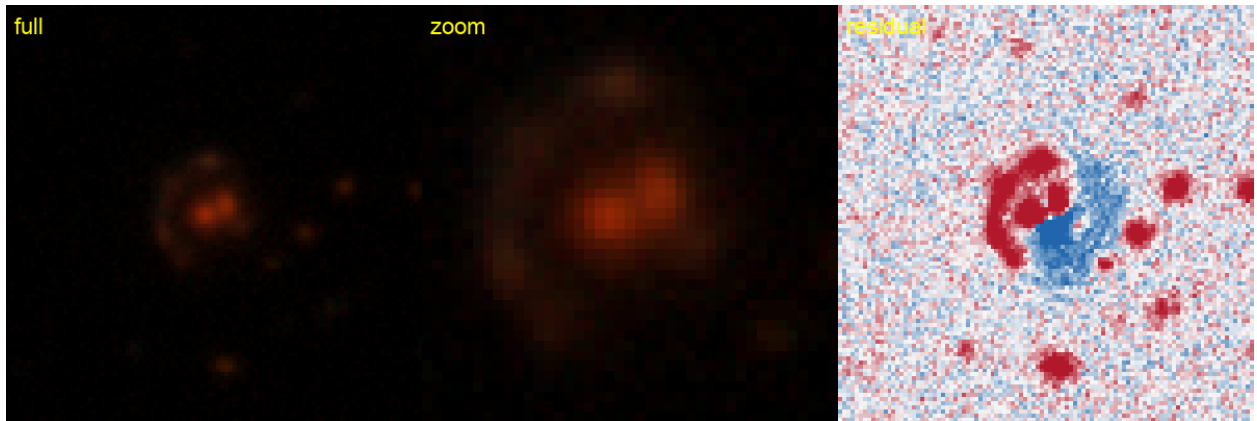


Figure 50: #57 s_91761_4679 — RA 350.3683, Dec -46.5138; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.86.
KNOWN — DES: DES J232128.37-463049.3, 0.25'' (SIMBAD).

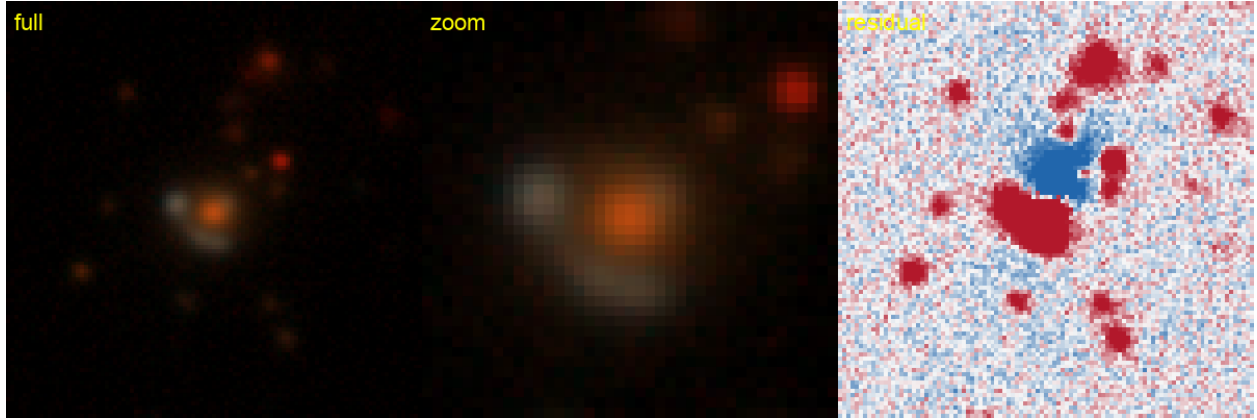


Figure 51: #59 s_107312.5223 — RA 23.8451, Dec -42.5398; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.85.
KNOWN — DES: [DBL2017] DES J0135-4232 (A), 0.18'' (SIMBAD).

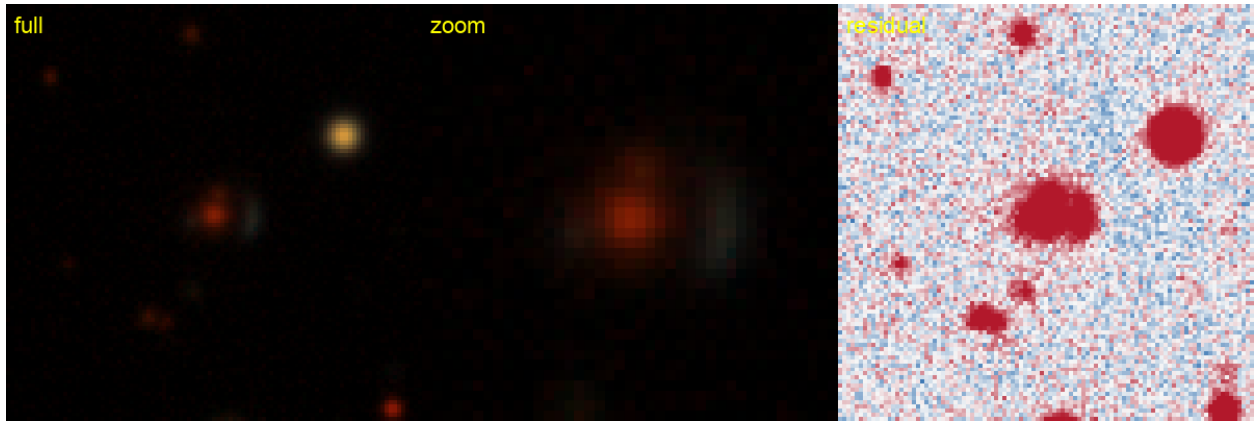


Figure 52: #60 s_69087.268 — RA 307.2325, Dec -52.5218; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.81.
KNOWN — DES: DES J202855.79-523118.3, 0.12'' (SIMBAD).

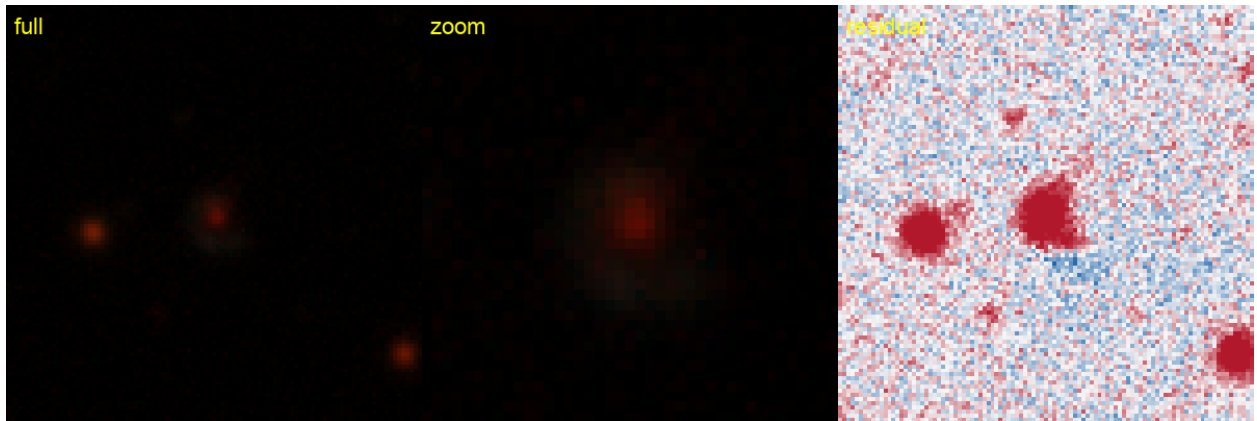


Figure 53: #61 s_165468.9648 — RA 64.7771, Dec -29.9937; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.78.
KNOWN — DES: DES J041906.50-295937.7, 0.49'' (SIMBAD).

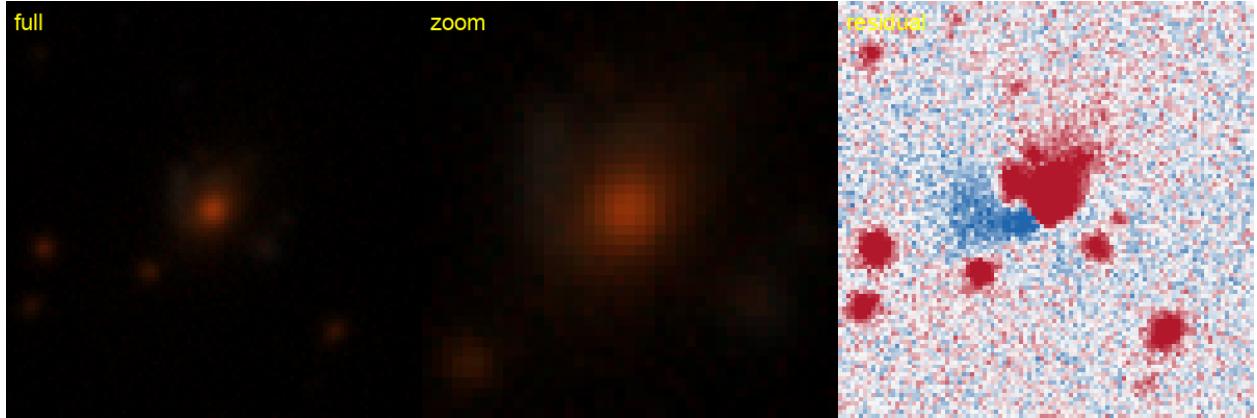


Figure 54: #62 s_256199_9097 — RA 29.6819, Dec -13.0258; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.78.
KNOWN — DES: DES J015843.66-130132.8, 0.18'' (SIMBAD).

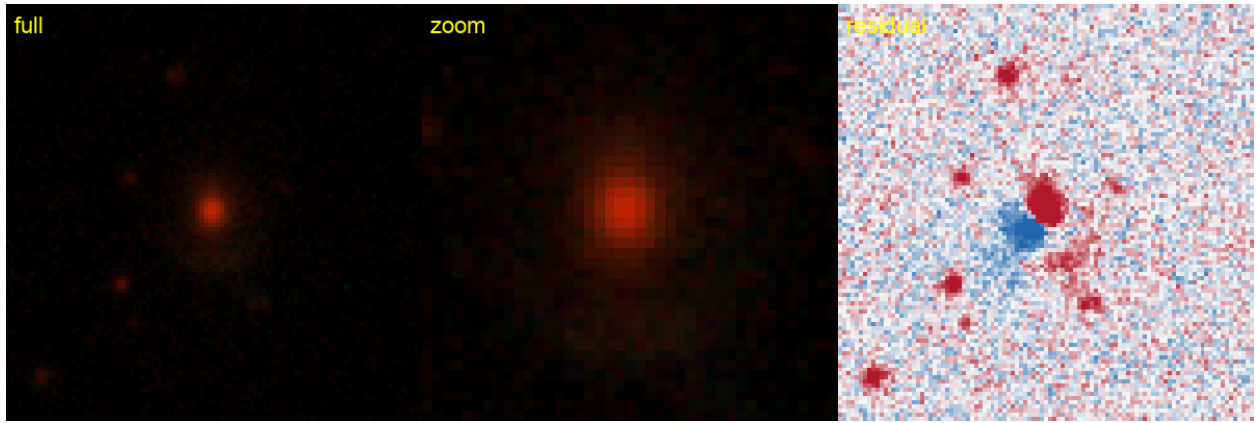


Figure 55: #63 s_364017_5199 — RA 141.0626, Dec +5.7690; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.77.
KNOWN — Huang+2020: DESI-141.0626+05.7690, 0.11'' (lineage).

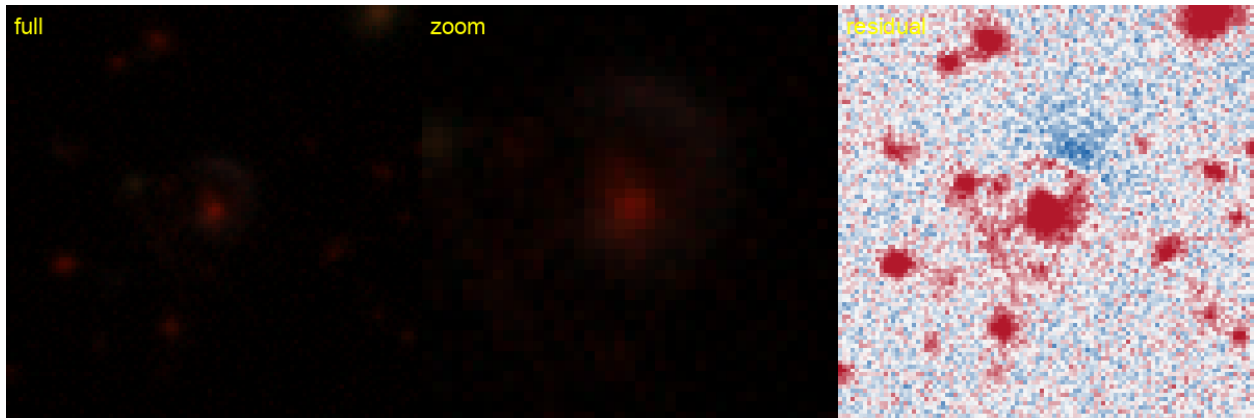


Figure 56: #66 s_119342_4136 — RA 50.7164, Dec -39.7694; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.90.
KNOWN — DES: DES J032251.91-394609.6, 0.17'' (SIMBAD).

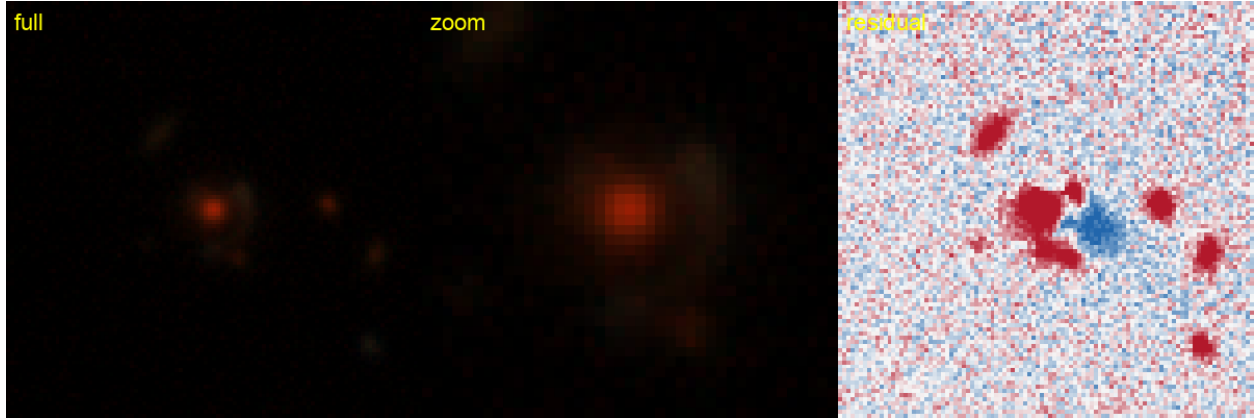


Figure 57: #68 s.209576.5226 — RA 65.5760, Dec -21.5461; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.86.
KNOWN — DES: DES J042218.21-213245.8, 0.28'' (SIMBAD).

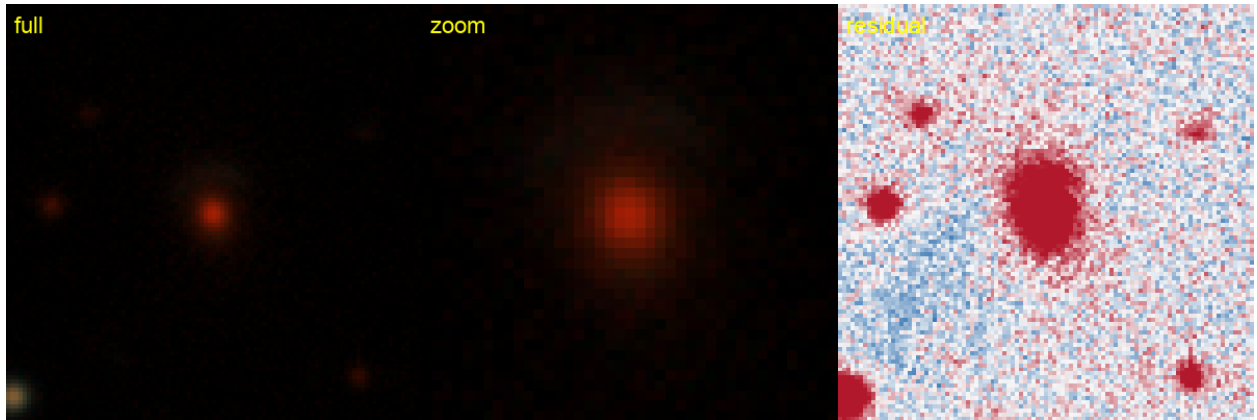


Figure 58: #69 s.225679.7025 — RA 25.7203, Dec -18.5211; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.79.
KNOWN — AGEL: AGEL J014253-183116 system, 0.94'' (SIMBAD).

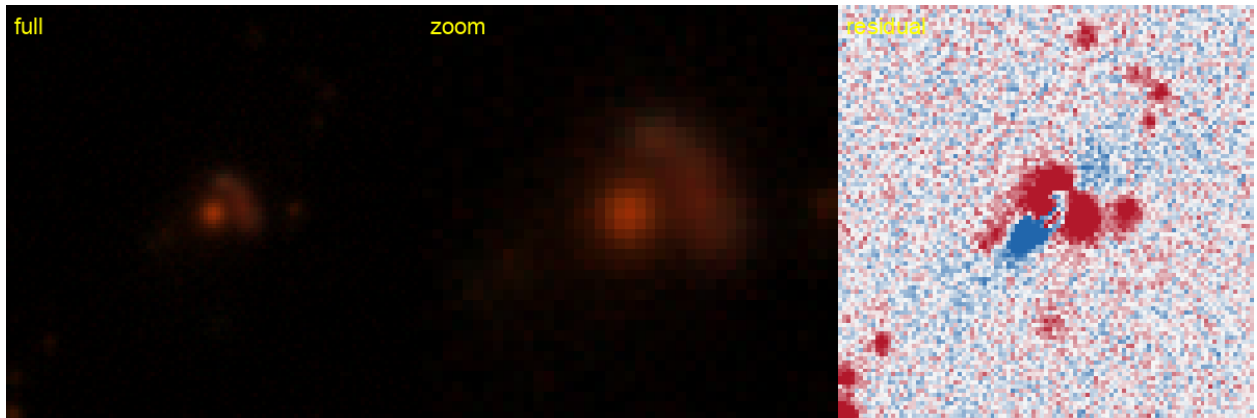


Figure 59: #73 s.129372.3768 — RA 16.4504, Dec -37.4286; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.82.
KNOWN — DES: DES J010548.04-372542.4, 0.71'' (SIMBAD).

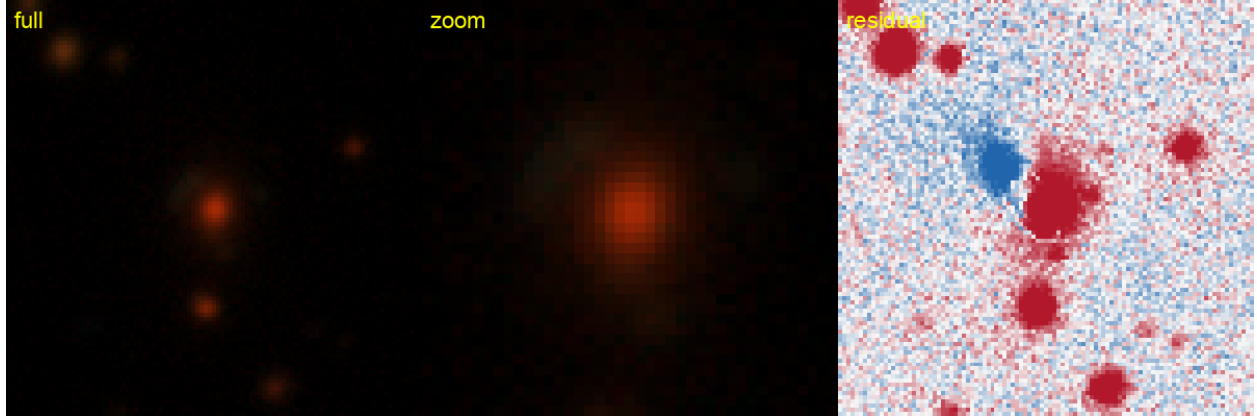


Figure 60: #74 s.238185_2253 — RA 58.5761, Dec -16.1645; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.82.
KNOWN — AGEL: AGEL J035418-160952 system, 0.51'' (SIMBAD).

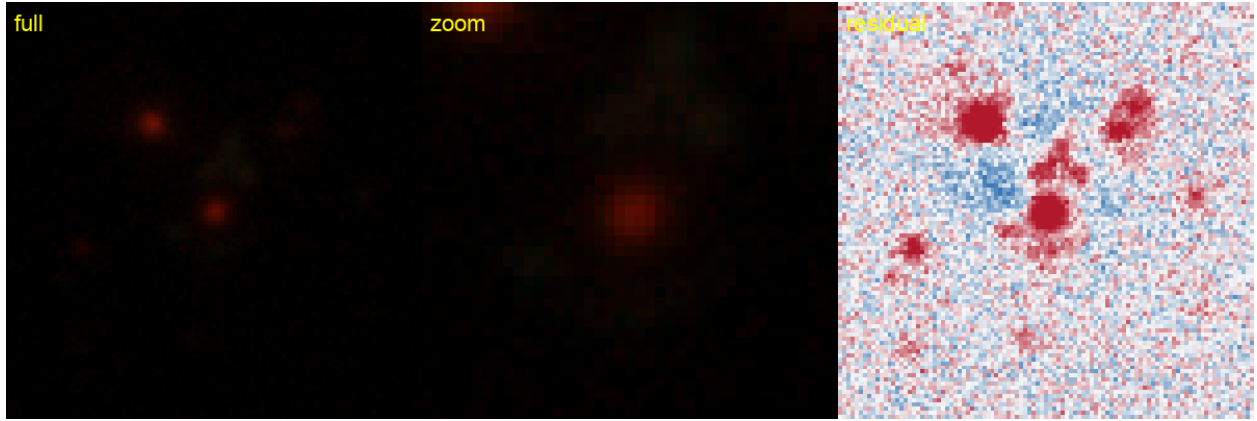


Figure 61: #75 s.94933_7964 — RA 47.5548, Dec -45.5743; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.78.
KNOWN — DES: [DBL2017] DES J0310-4534 (A), 0.14'' (SIMBAD).

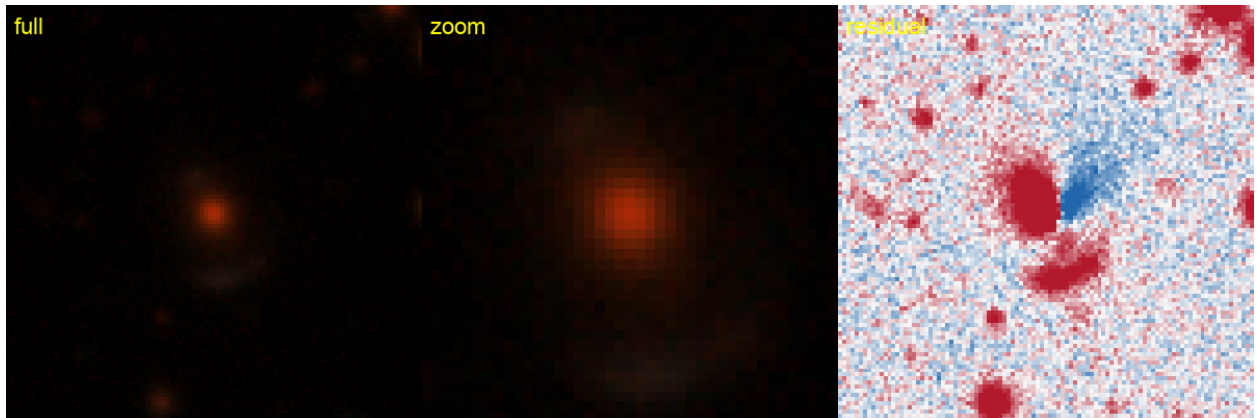


Figure 62: #77 s.71871_246 — RA 353.9664, Dec -51.8716; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.76.
KNOWN — AGEL: AGEL J233552-515218 system, 0.95'' (SIMBAD).

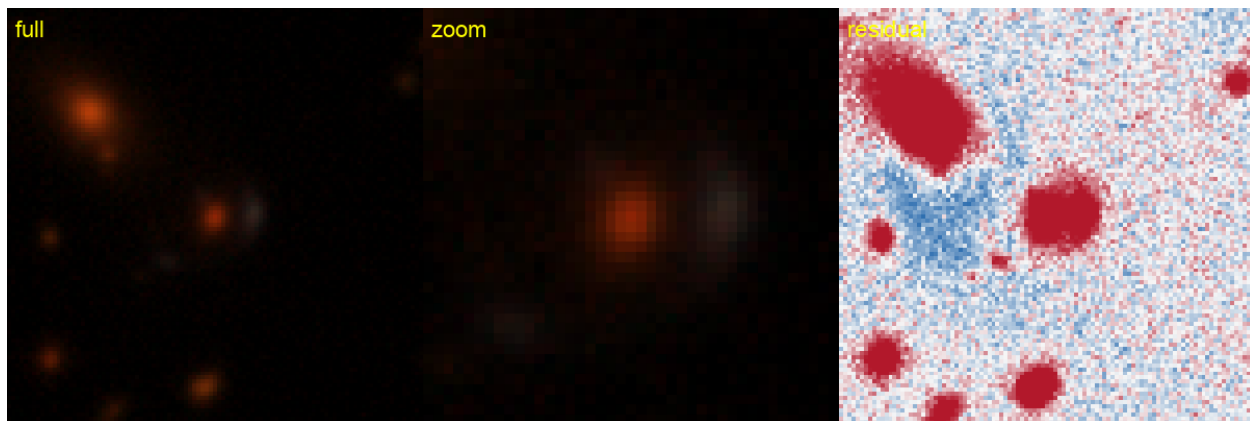


Figure 63: #78 s_90920_9419 — RA 45.9517, Dec -46.4407; grade **B**, tier-1 p_{lens} 0.30, v3blend8 0.85.
KNOWN — DES: [DBL2017] DES J0303-4626 (A), 0.15'' (SIMBAD).

New — candidates new to the literature (25 of 78)

These are absent from the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature; all are grade B, and at DECaLS 1'' an unknown (likely large) fraction are lrg+companion mimics. Ranked by tier-1 p_{lens} .

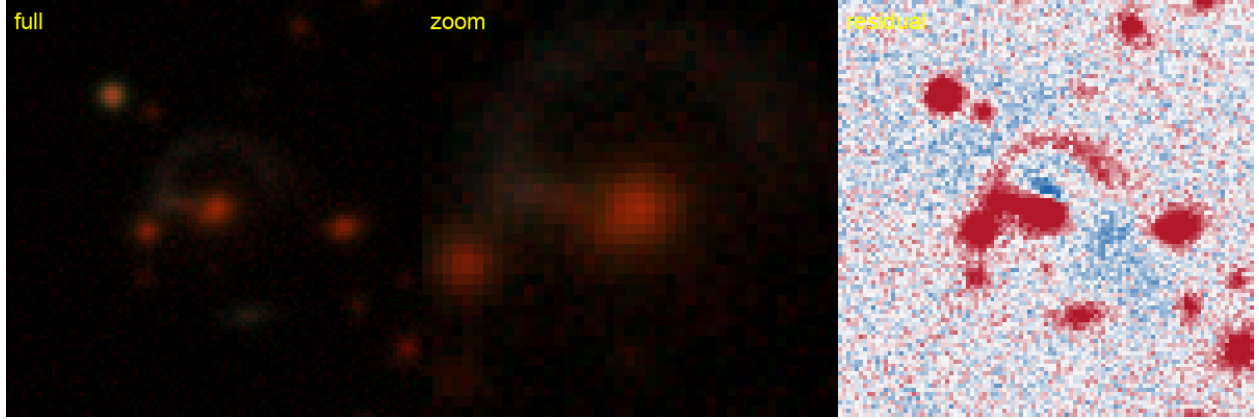


Figure 64: #9 s_230539_4159 — RA 222.0715, Dec -17.8489; grade **B**, tier-1 p_{lens} 0.72, v3blend8 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

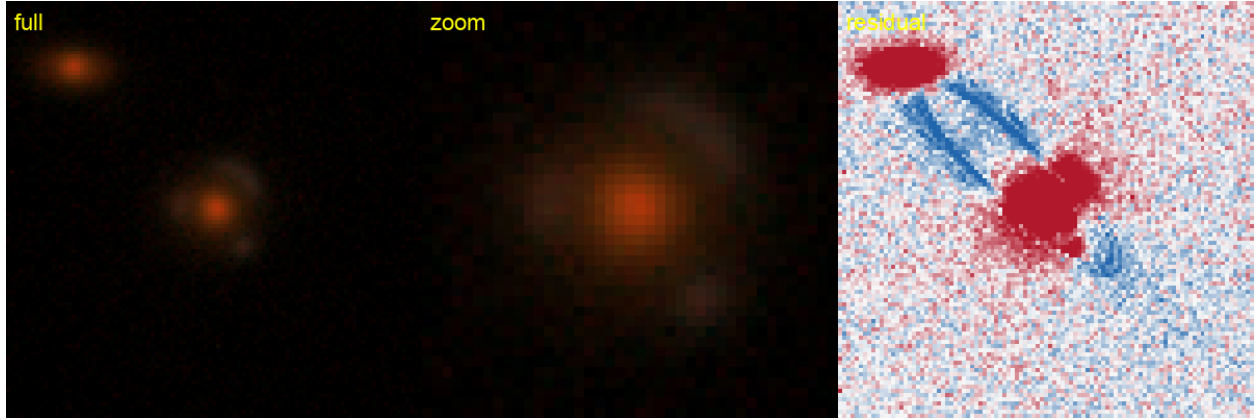


Figure 65: #13 s_234689_542 — RA 227.0427, Dec -16.8771; grade **B**, tier-1 p_{lens} 0.62, v3blend8 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

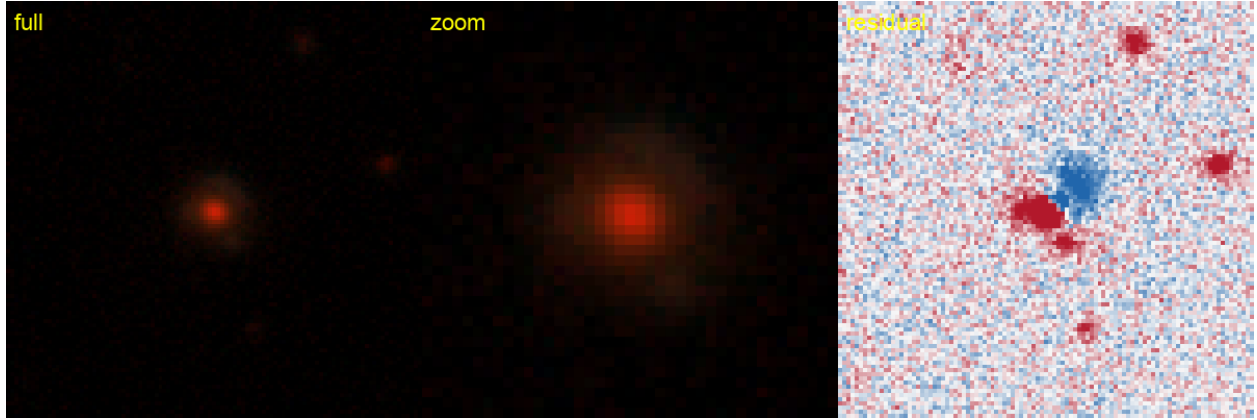


Figure 66: #15 s_205319_6061 — RA 359.9772, Dec -22.4596; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.88. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

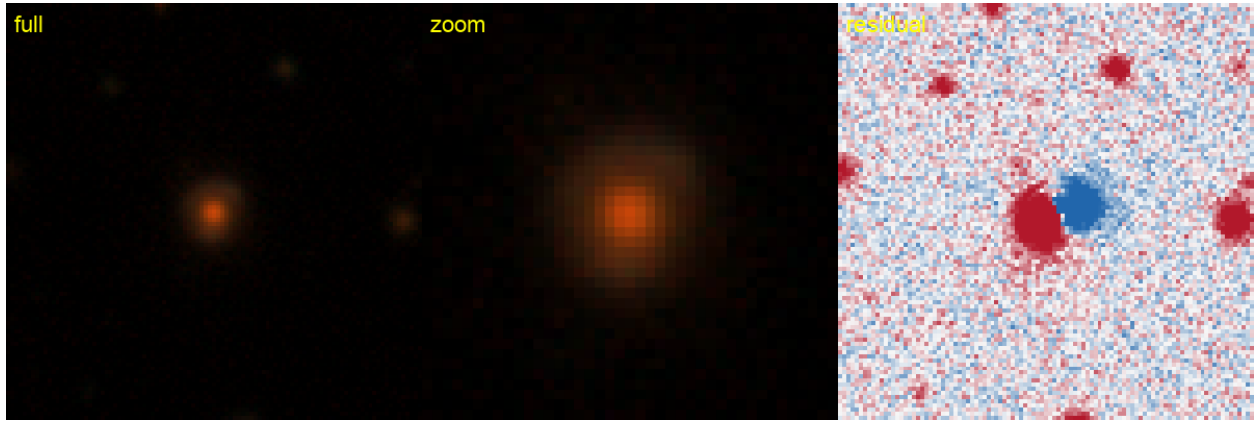


Figure 67: #18 s_218895_5088 — RA 32.7525, Dec -19.6362; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.81. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

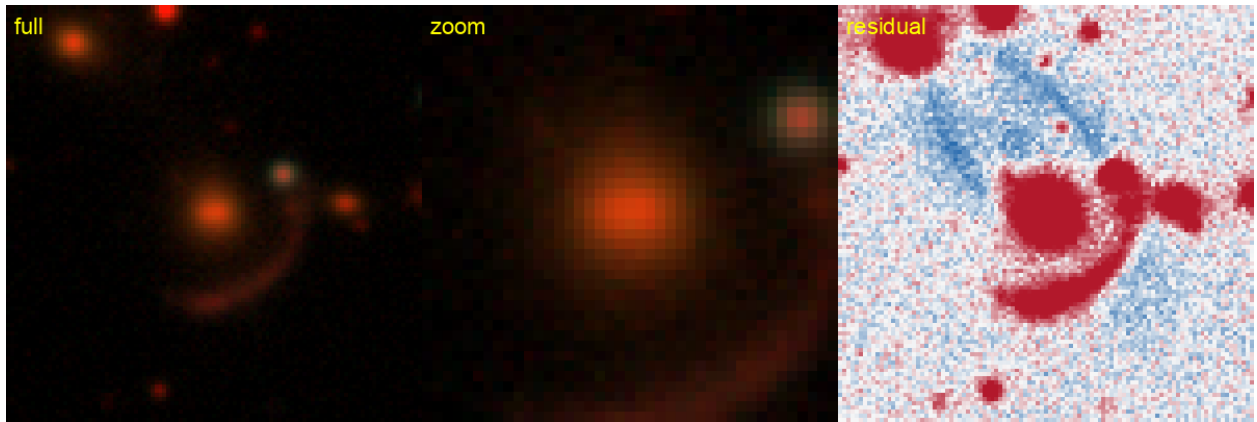


Figure 68: #22 s_194300_2796 — RA 242.6052, Dec -24.4155; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

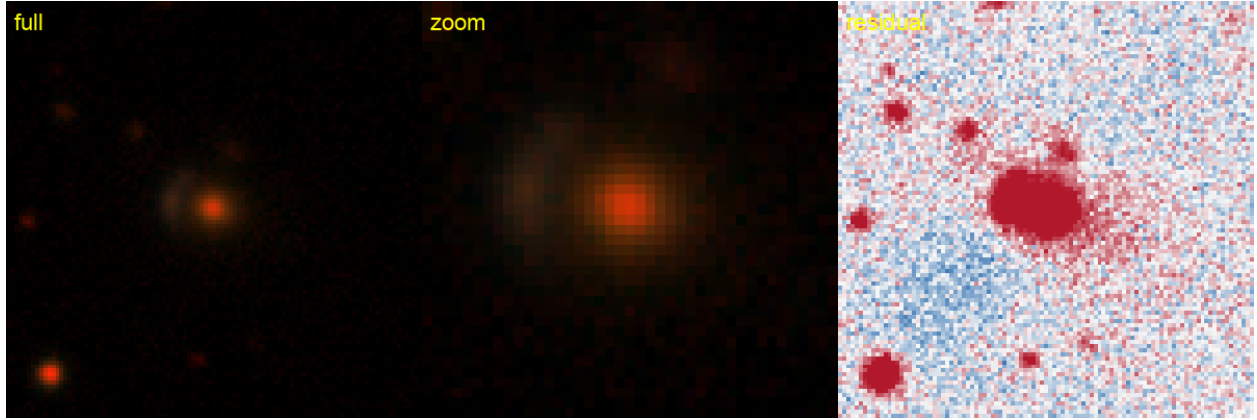


Figure 69: #28 s.23764.4161 — RA 132.6603, Dec -68.1876; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

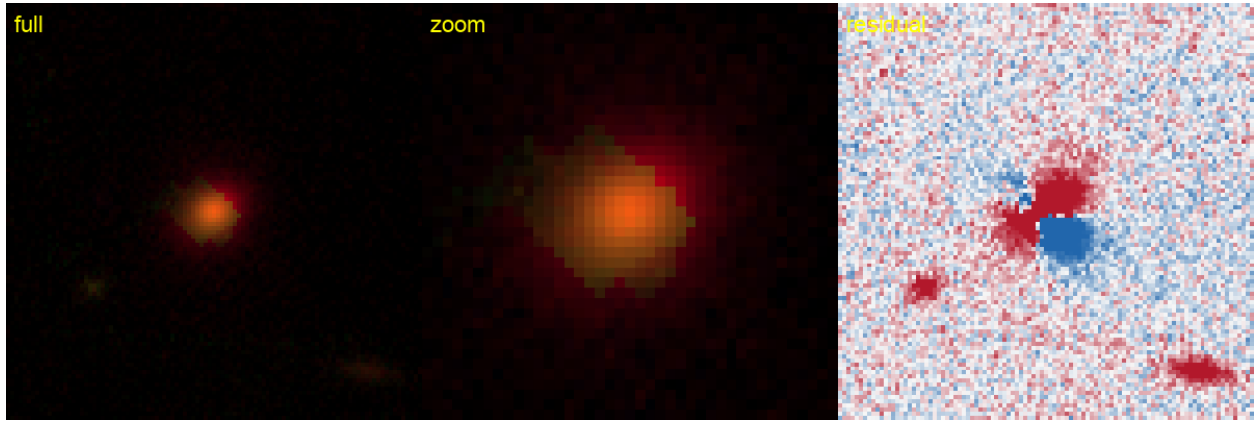


Figure 70: #32 s.353585.141 — RA 45.8222, Dec +3.9613; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

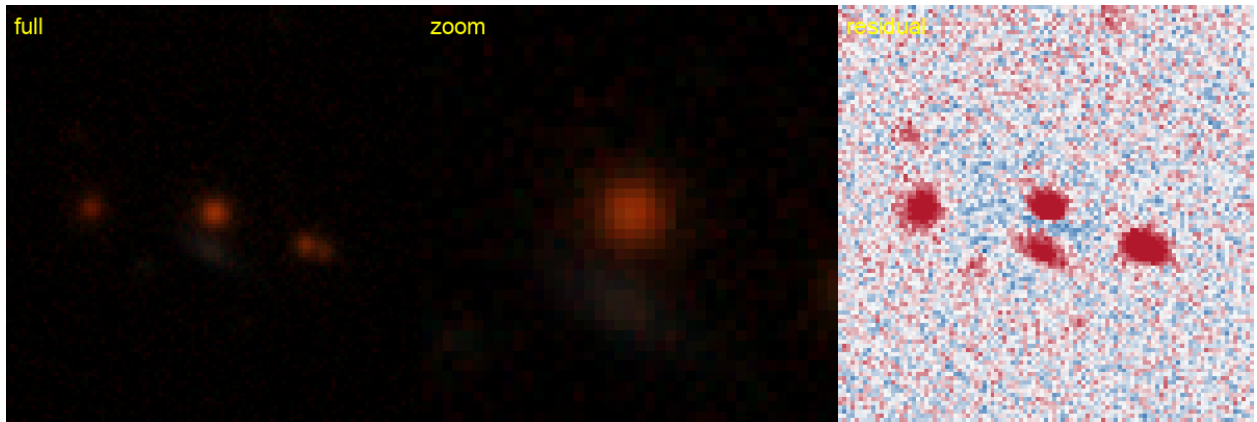


Figure 71: #35 s.289350.3716 — RA 164.8051, Dec -7.1449; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.81. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

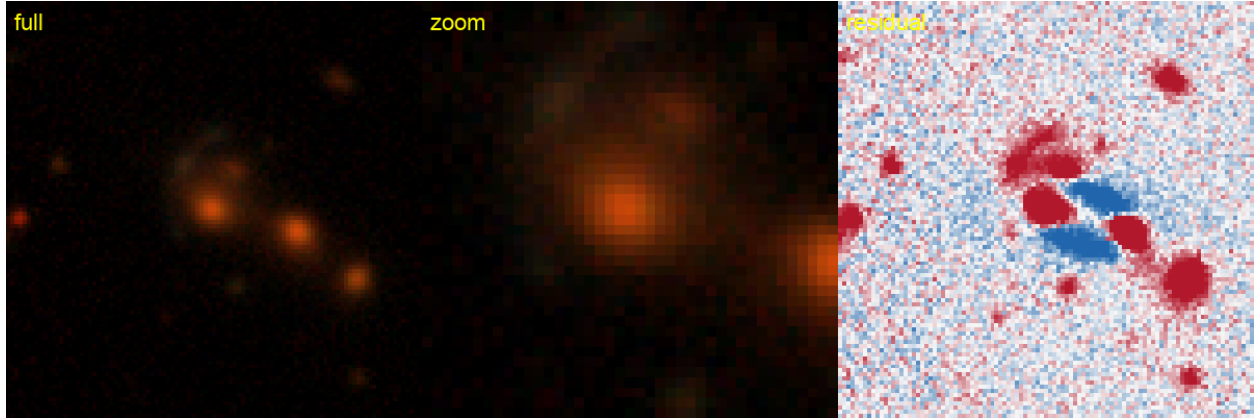


Figure 72: #36 s_148704_6732 — RA 183.8176, Dec -33.6006; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

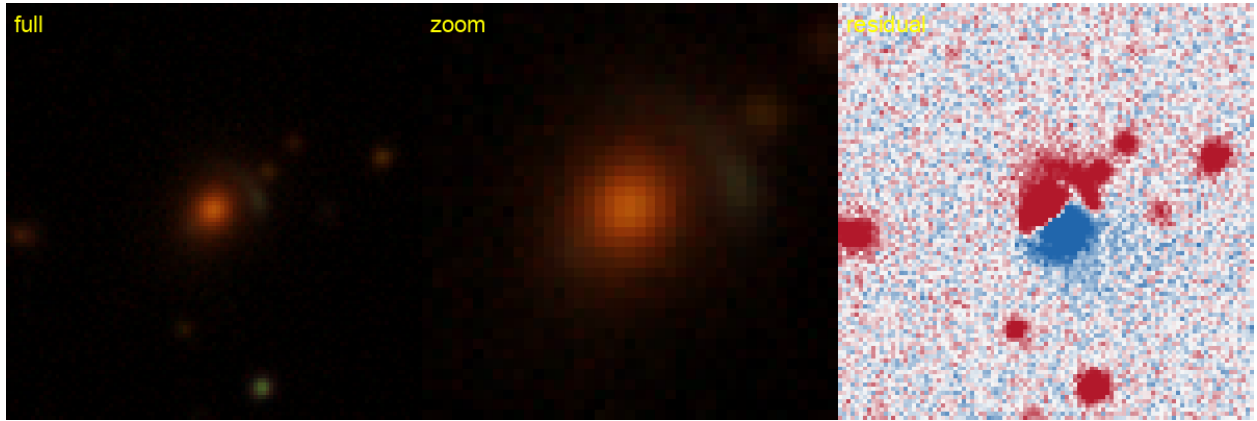


Figure 73: #41 s_90098_8002 — RA 107.1696, Dec -46.7832; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.83. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

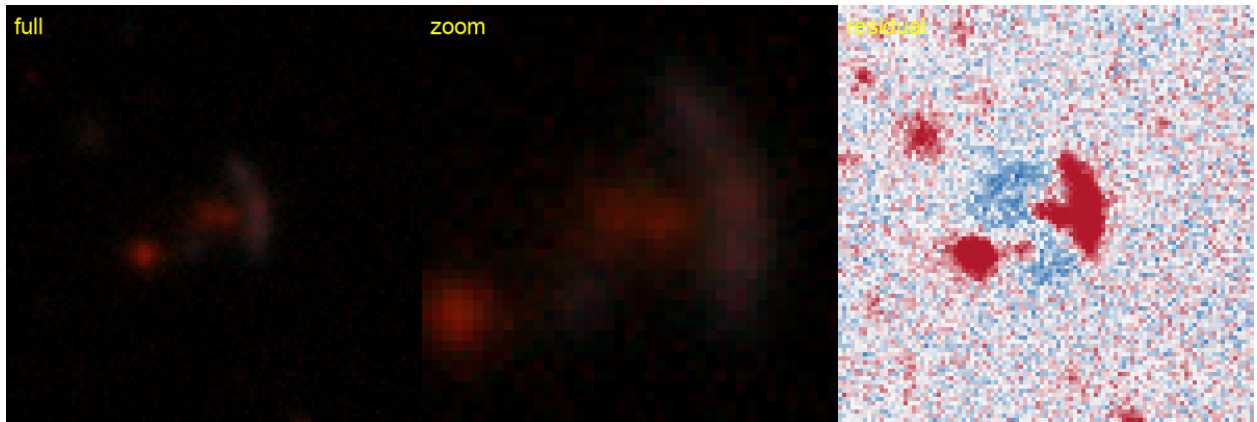


Figure 74: #43 s_502450_1680 — RA 161.1145, Dec +31.2344; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

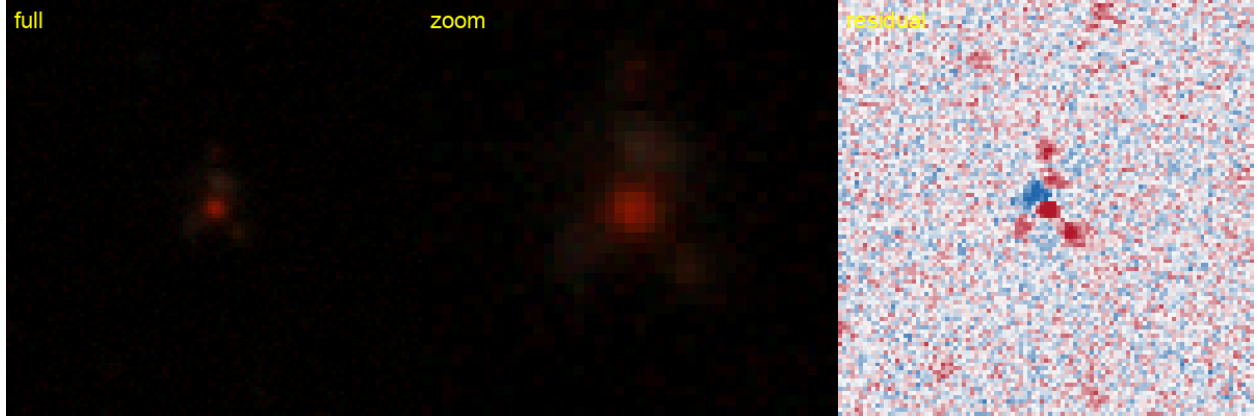


Figure 75: #44 s_426295_4804 — RA 182.0229, Dec +16.6326; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

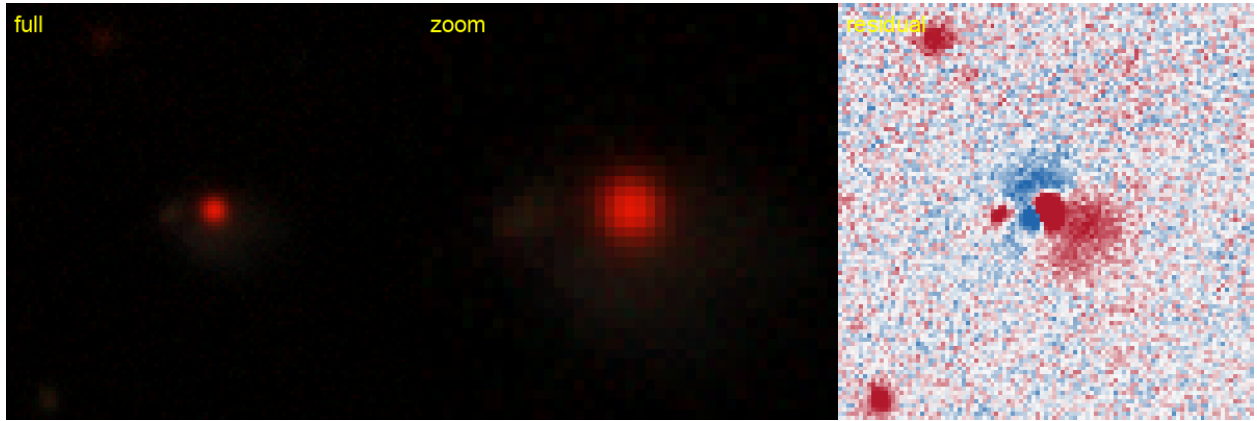


Figure 76: #45 s_369309_653 — RA 30.4465, Dec +6.6551; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

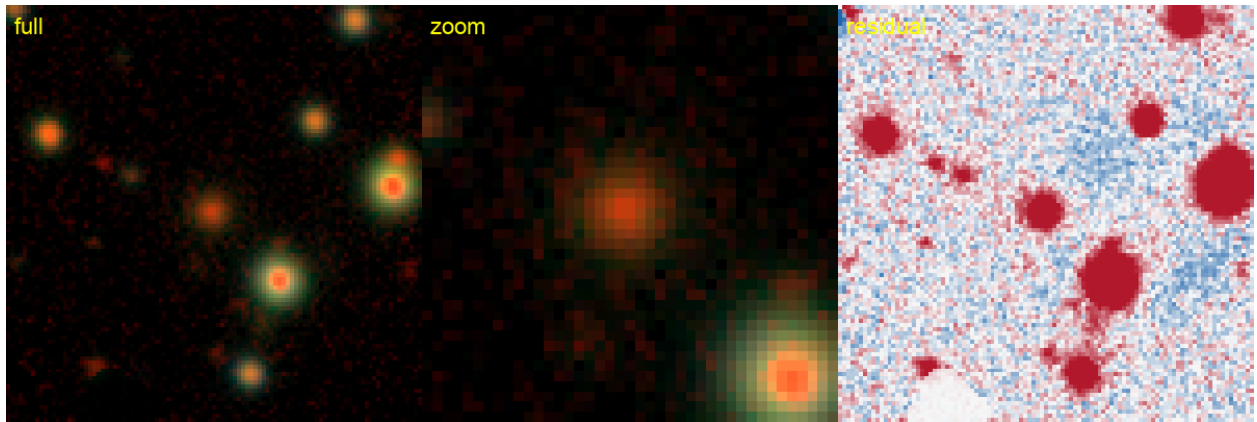


Figure 77: #46 s_32666_26056 — RA 248.6090, Dec -64.5569; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.73. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

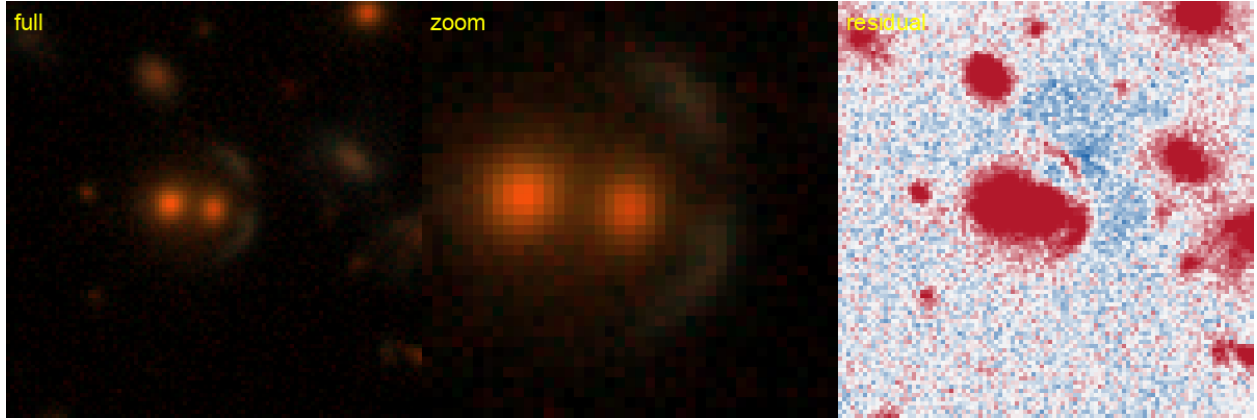


Figure 78: #52 s_266620_918 — RA 172.1320, Dec -11.3108; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

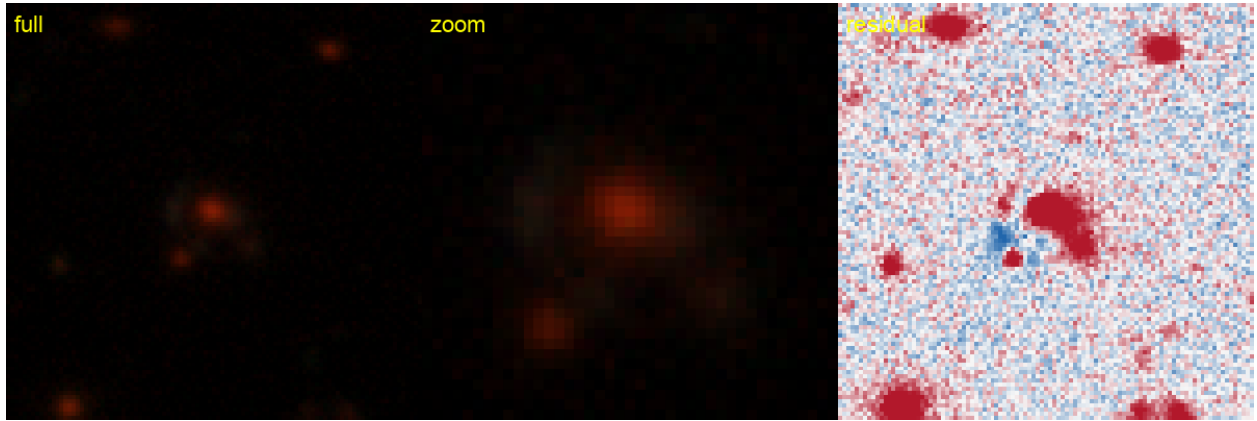


Figure 79: #53 s_242312_4967 — RA 50.3567, Dec -15.5010; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.90. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

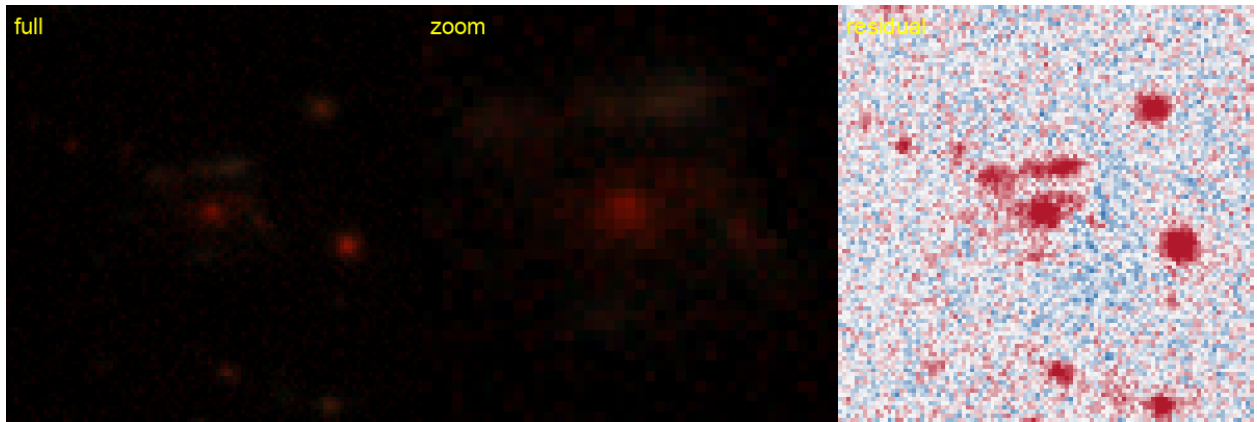


Figure 80: #56 s_262449_2840 — RA 188.7861, Dec -11.9798; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.87. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

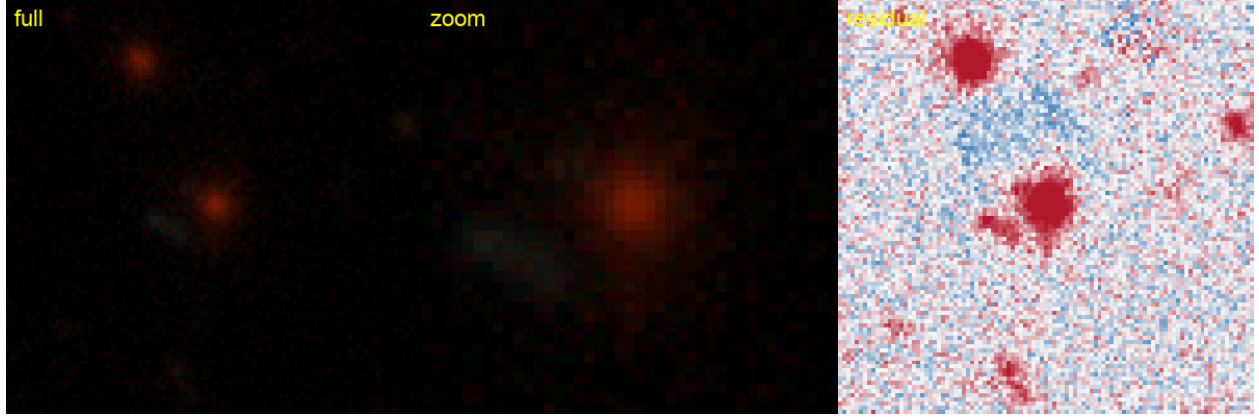


Figure 81: #58 s_360639_3636 — RA 13.4868, Dec +5.3590; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.85. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

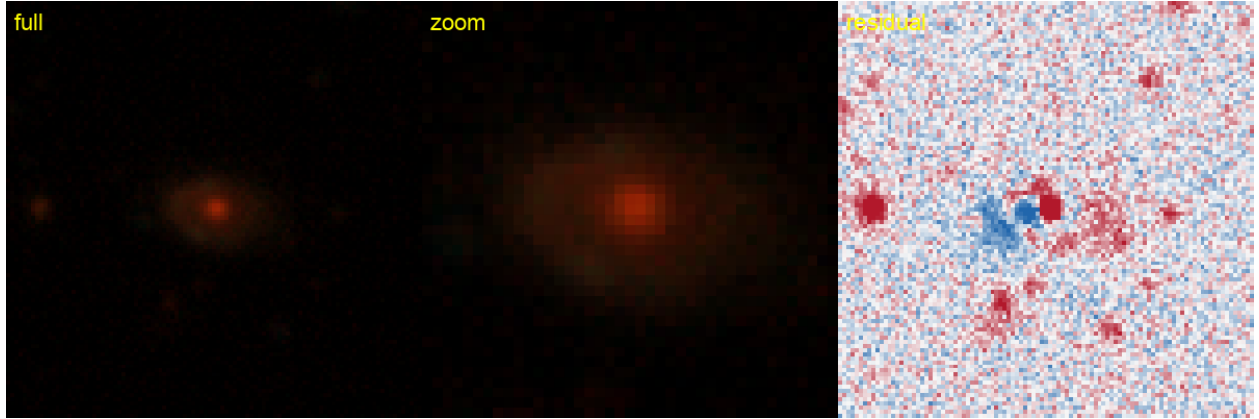


Figure 82: #64 s_188312_3322 — RA 33.2537, Dec -25.3945; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.76. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

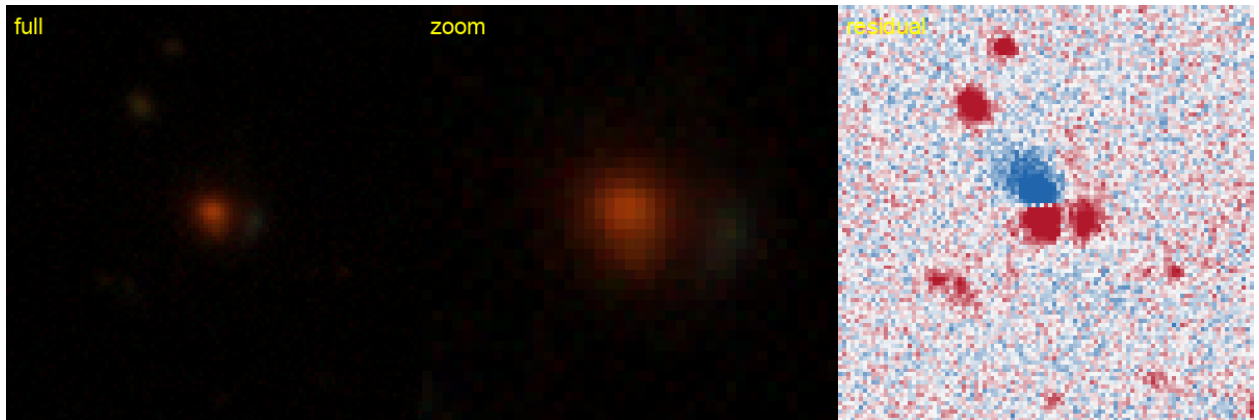


Figure 83: #65 s_188754_2468 — RA 155.4695, Dec -25.5029; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

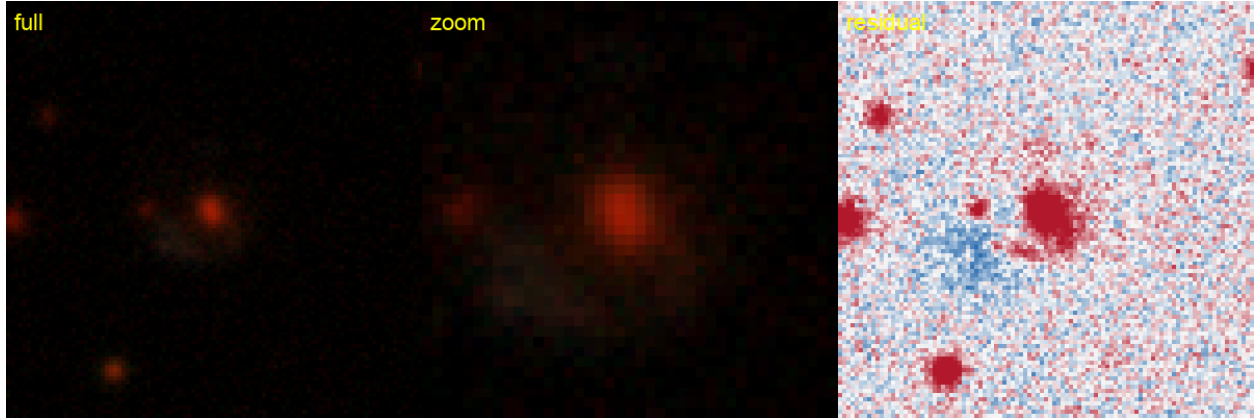


Figure 84: #67 s_405220_1292 — RA 136.4655, Dec +13.0307; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.89. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

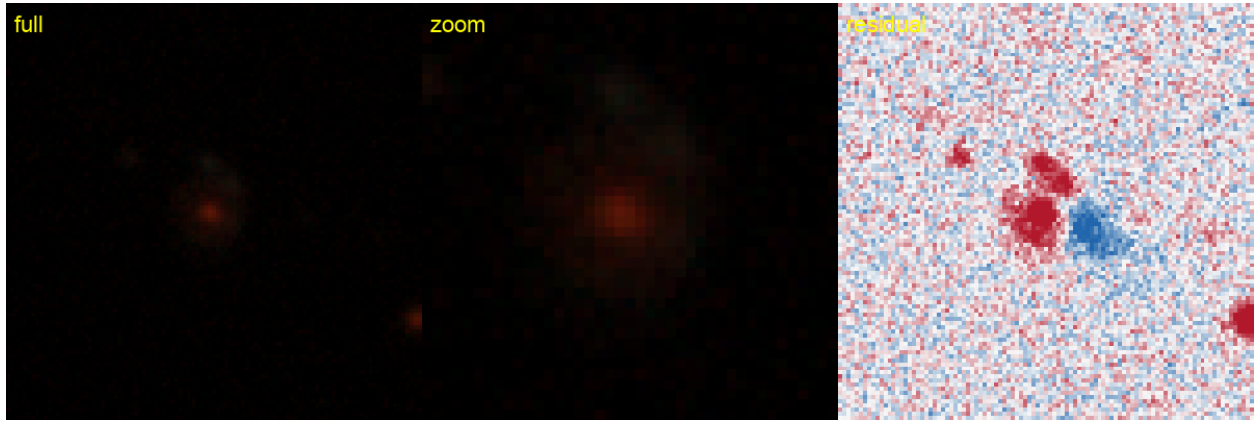


Figure 85: #70 s_250611_2161 — RA 33.7898, Dec -14.1185; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.77. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

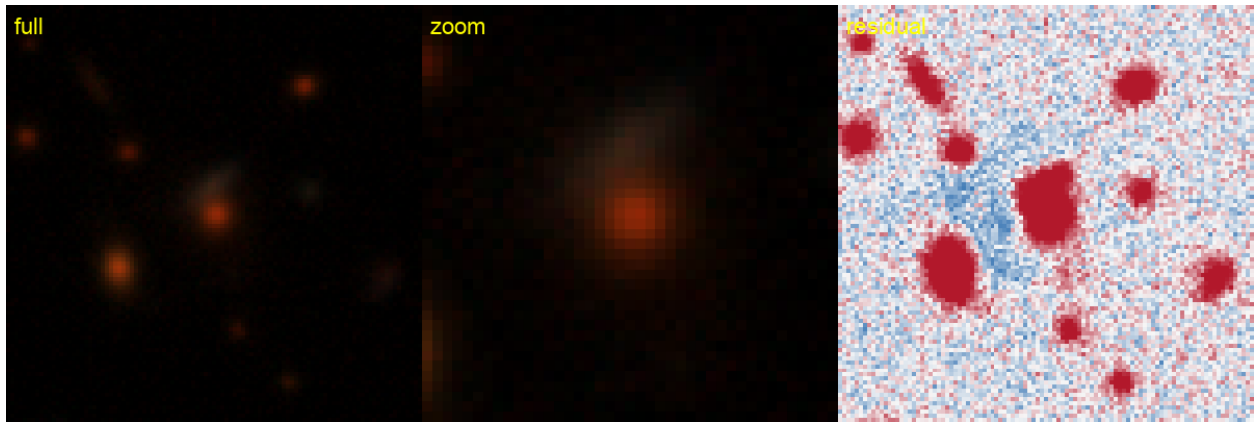


Figure 86: #71 s_161725_5181 — RA 61.4085, Dec -30.6887; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.76. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

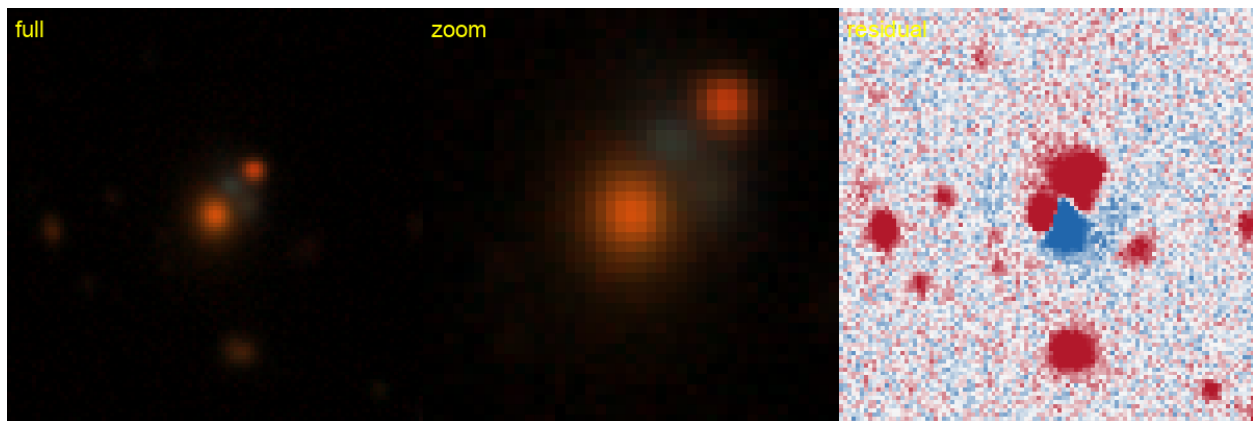


Figure 87: #72 s_210799_616 — RA 33.4979, Dec -21.2827; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.75. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

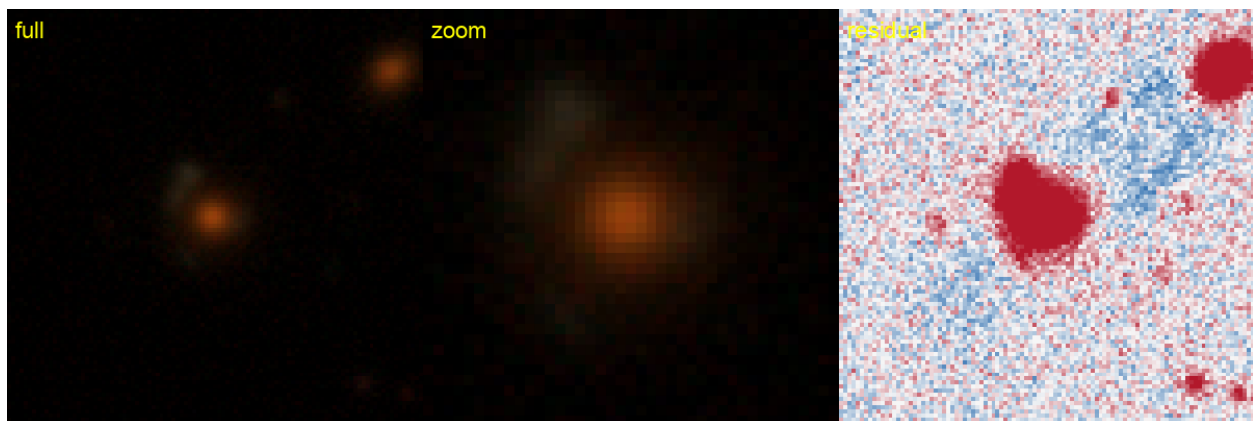


Figure 88: #76 s_250997_4626 — RA 133.2708, Dec -13.8944; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.77. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

Table 5: The 24 confirmed tier-2 grade-A/B systems tracked across the three residual arms (grade / p_{lens}). **All 9/9 new-class campaign discoveries survive as A/B under the honest residual**; 13/15 SuGOHI cross-matches do likewise. $\downarrow$ marks the two SuGOHI lenses the honest grader scores below A/B (one already lost to nondeterminism in the legacy-now control); both are independently-confirmed lenses, i.e. conservative grading, not campaign candidates.

name	class	prior	legacy-now	new (χ)
s_341950_6798	SuGOHI	A/0.97	A/0.95	A/0.96
s_340513_3688	NEW	A/0.97	A/0.95	A/0.97
s_328891_7514	SuGOHI	A/0.97	A/0.93	A/0.93
s_331753_3591	SuGOHI	A/0.95	A/0.97	A/0.97
s_336034_1750	SuGOHI	A/0.95	A/0.97	A/0.97
s_336034_1788	SuGOHI	A/0.95	A/0.95	A/0.95
s_318982_1451	SuGOHI	A/0.93	A/0.93	A/0.93
s_327526_9059	NEW	A/0.92	A/0.88	A/0.80
s_351303_461	NEW	A/0.92	A/0.85	A/0.88
s_345385_1515	NEW	A/0.92	A/0.90	A/0.92
s_323786_3896	SuGOHI	A/0.91	D/0.04	A/0.88
s_327403_2882	SuGOHI	A/0.91	A/0.90	A/0.92
s_340971_2860	NEW	A/0.91	A/0.88	A/0.85
s_345385_1514	NEW	A/0.87	A/0.88	A/0.87
s_351084_1943 $\downarrow$	SuGOHI	A/0.87	A/0.83	C/0.22
s_332668_5796	SuGOHI	A/0.87	B/0.72	A/0.83
s_326089_1785	NEW	A/0.82	D/0.04	B/0.65
s_325163_2981	NEW	A/0.82	A/0.82	A/0.82
s_300291_2747	SuGOHI	A/0.82	A/0.75	A/0.87
s_325489_4248	SuGOHI	B/0.68	A/0.80	B/0.58
s_329032_842	SuGOHI	B/0.65	B/0.58	A/0.78
s_336801_4965	SuGOHI	B/0.65	D/0.05	B/0.65
s_318043_1037	NEW	B/0.45	B/0.40	B/0.48
s_346996_4322 $\downarrow$	SuGOHI	B/0.40	D/0.03	C/0.08

C Residual-robustness A/B: the confirmed catalogue across residual arms

Table 5 tracks all 24 confirmed tier-2 grade-A/B systems (§3: the 9 new-class detections of Table 2 plus the 15 SuGOHI cross-matches) across the three arms of the paired residual re-run (§3, “Residual robustness”): the stored legacy vetting run, a legacy-residual rerun (the LLM-nondeterminism control), and the new honest- χ run. All 9 new-class campaign discoveries survive as A/B under the honest residual. Reproduced by `compare_residual_ab.py` from the three cascade parquets.